EEE 201

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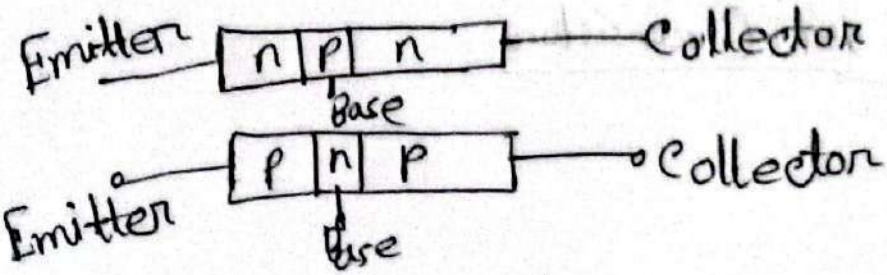
- BJT (Bipolar Junction Transistor)
- FET (Field Effect Transistor)

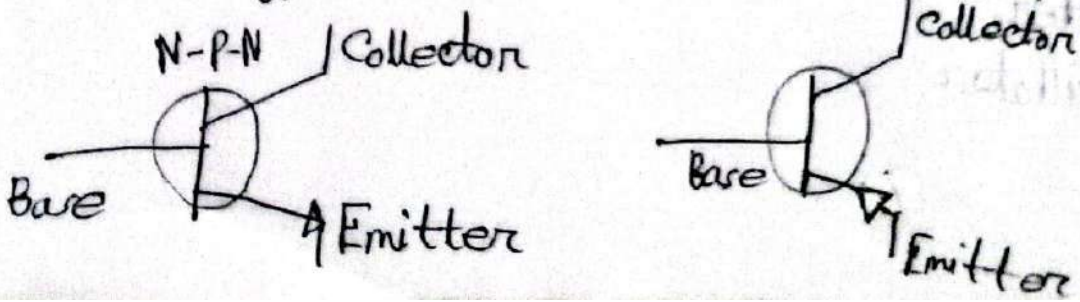
Transfer Varistor থেকে Transistor নামের উৎপত্তি।

- A transistor is a device that acts depending on the voltage level of the input signal, either as a conductor or as a resistor of the flow of electricity.

Conductor, Semiconductor, Isolator

BJT — PNP Transistor
NPN Transistor

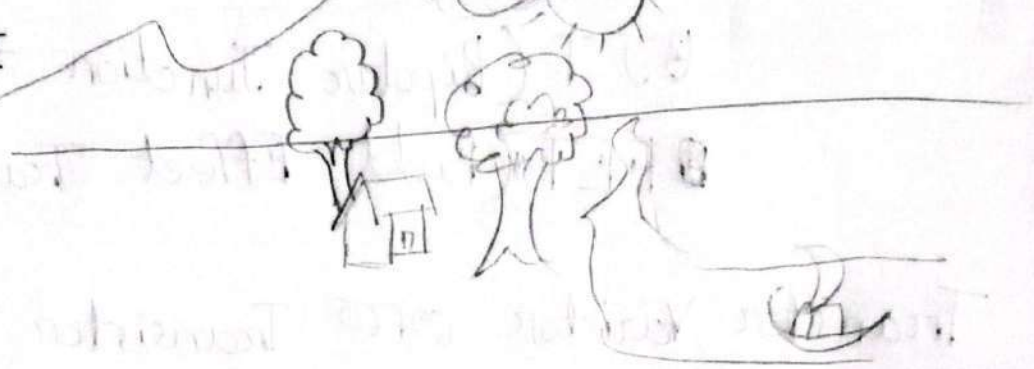




3/8/2026

Emitter: It is a region situated in one side of a a transistor, which supplies charge carriers to the other two regions

Base: I



Bipolar

Both polarity electrons and holes Participate in the conduction process

Junction

It has two P-N junction (CB & EB)

Transistor

It has transfer resistor. Its resistance changes with respect to region of operation.

Application of BGT (Transistor

1. Amplifier
2. Switch
3. Filter
4. Oscillator

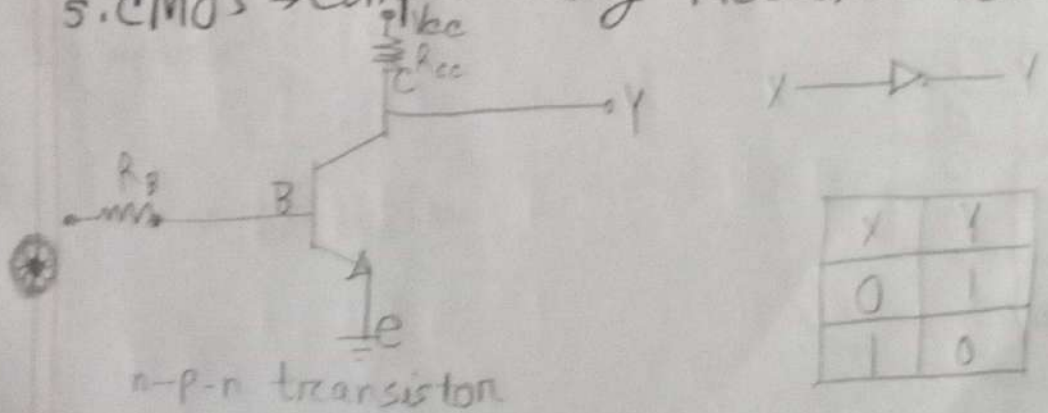
Advantages of BGT (Transistor)

1. High Voltage gain
2. Lower voltage supply
3. Lower power dissipation
4. Small size

Disadvantages of BGT (Transistor)

1. Temparature dependence
2. Lower switching speed

1. RTL → Resistor Transistor Logic
2. DTL → Diode Transistor Logic
3. TTL → Transistor Transistor Logic
4. NMOS → N-Channel Metaloxide Semi-conductor
5. CMOS → Complementary Metaloxide Semi Conductor (NMOS + PMOS)

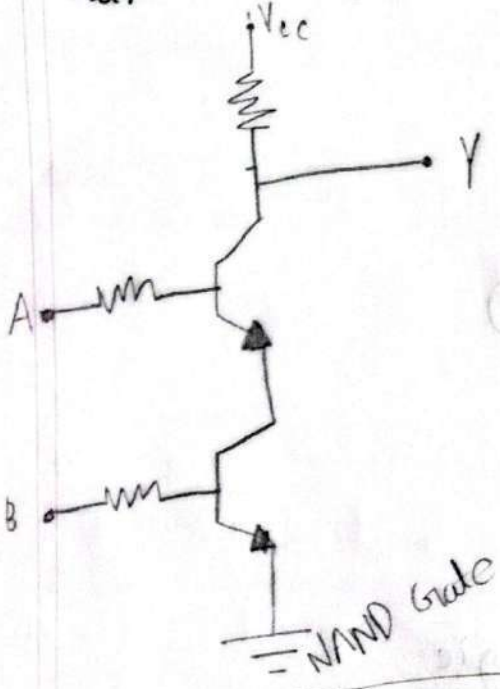


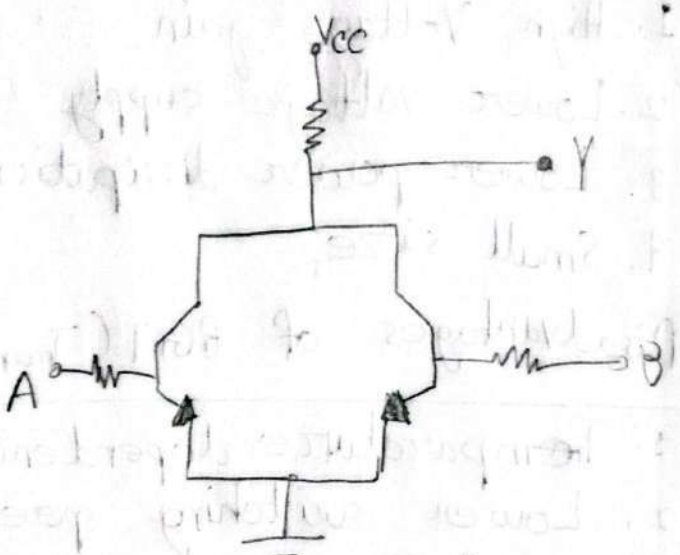
X	Y
0	1
1	0

NOT-Gate

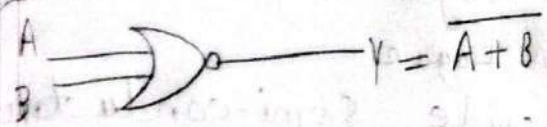
Product → Series

Sum → Parallal



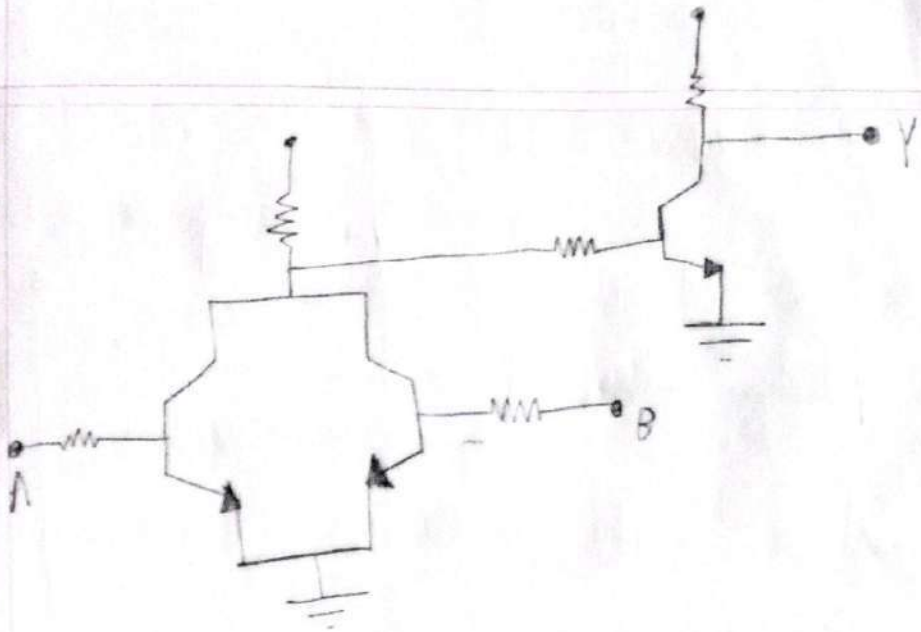


NOR Gate RTL

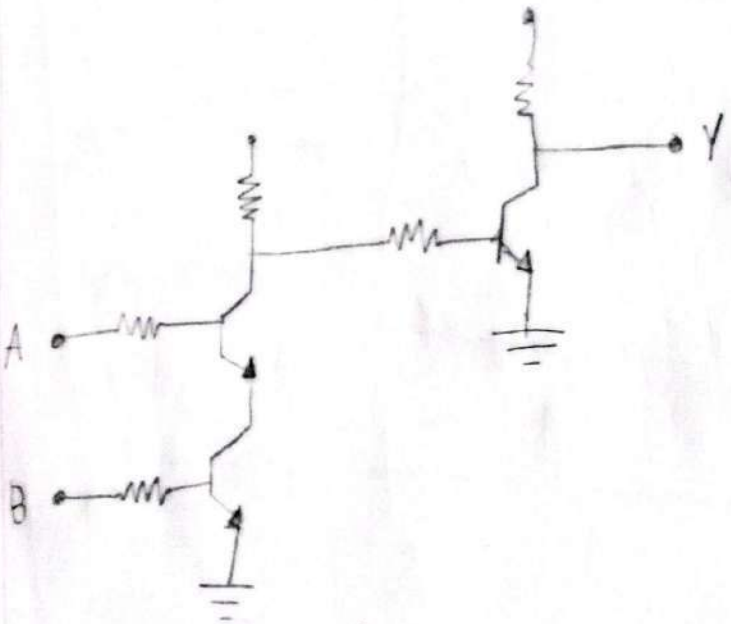


A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

NOR Gate



OR Gate



AND Gate

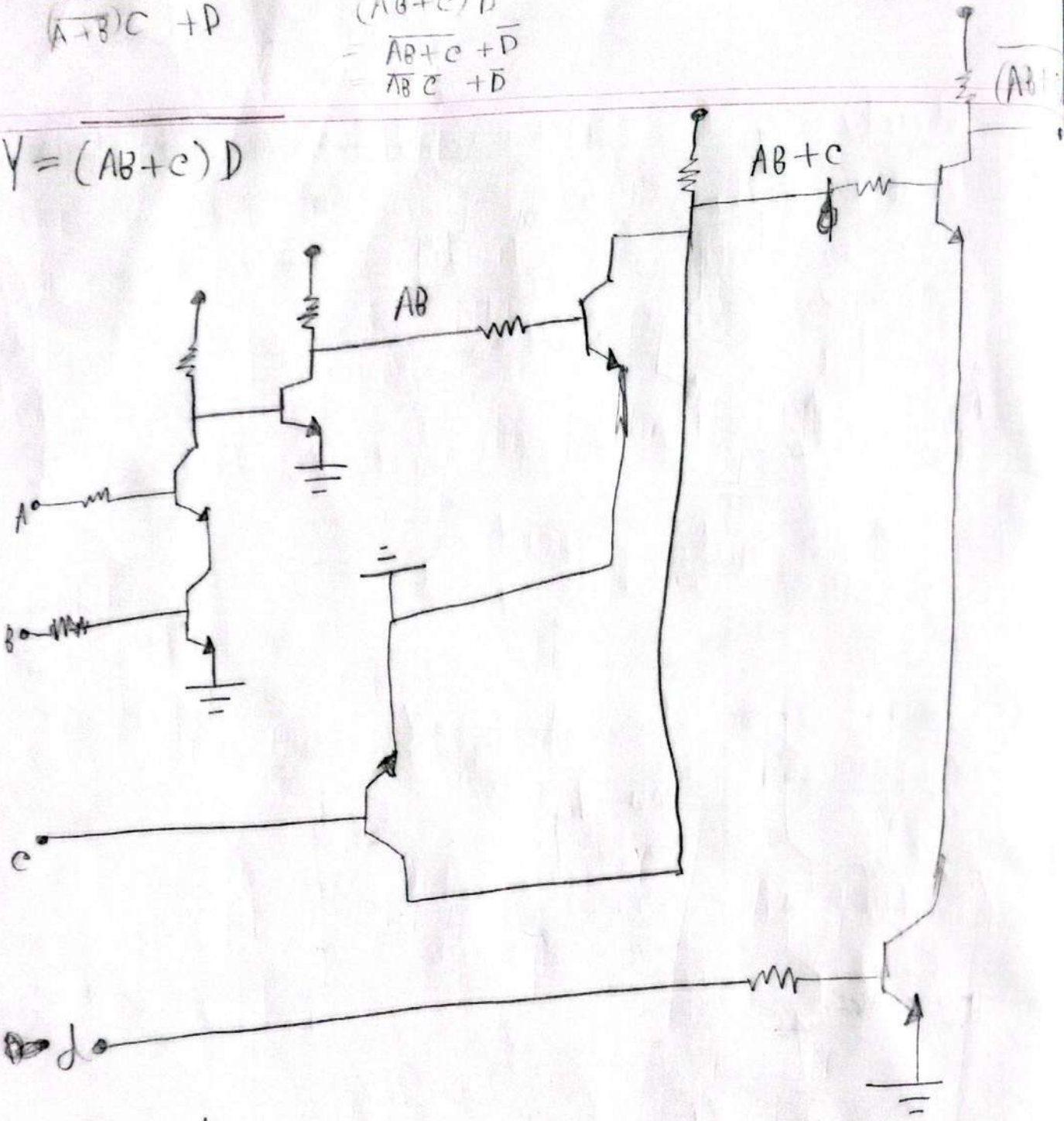
$\overline{(\overline{A+B})C + D}$

$(AB+C)D$

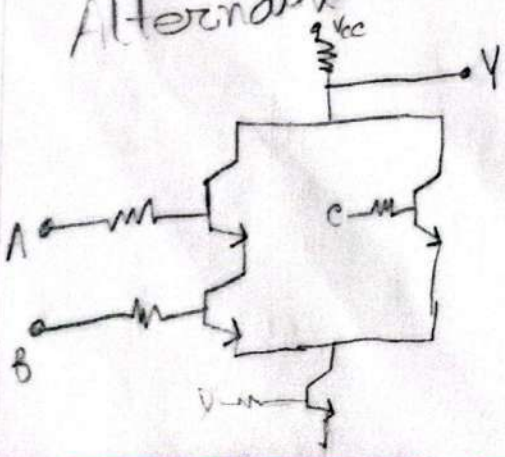
$= \overline{\overline{AB}+C} + \overline{D}$

$= \overline{AB}\,C + \overline{D}$

$Y = (AB+C)D$



Alternative



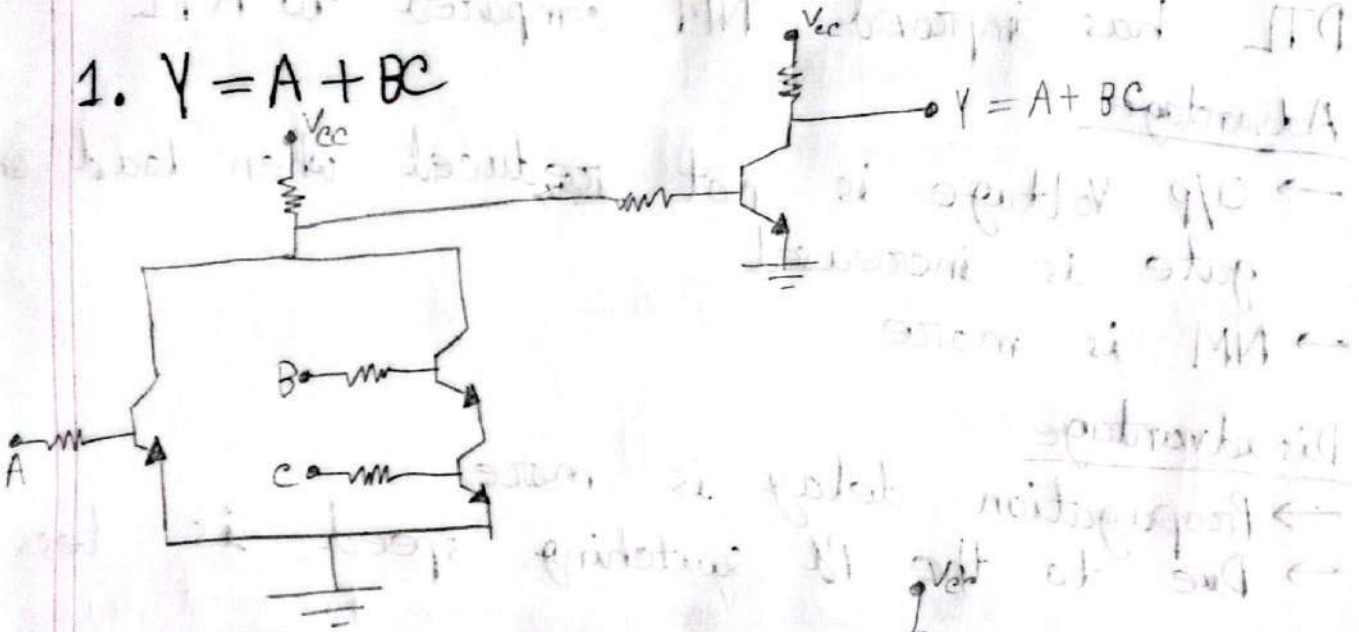
H.W

$Y = A + BC$ RTL Draw

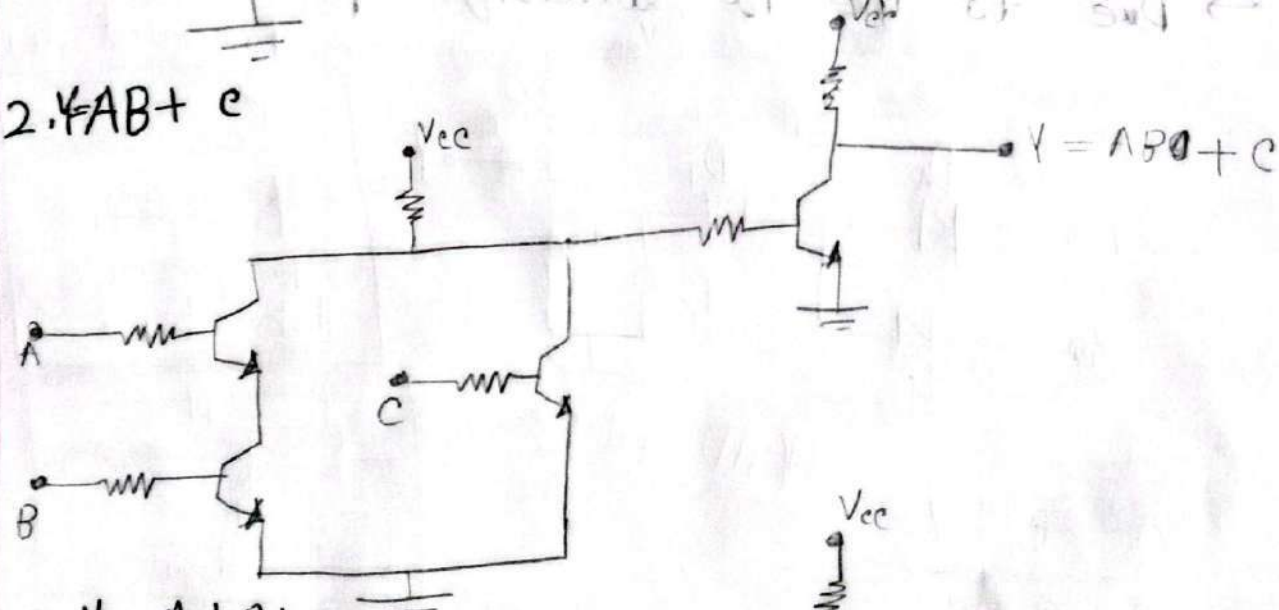
$Y = AB + C$

$Y = A + B + C$

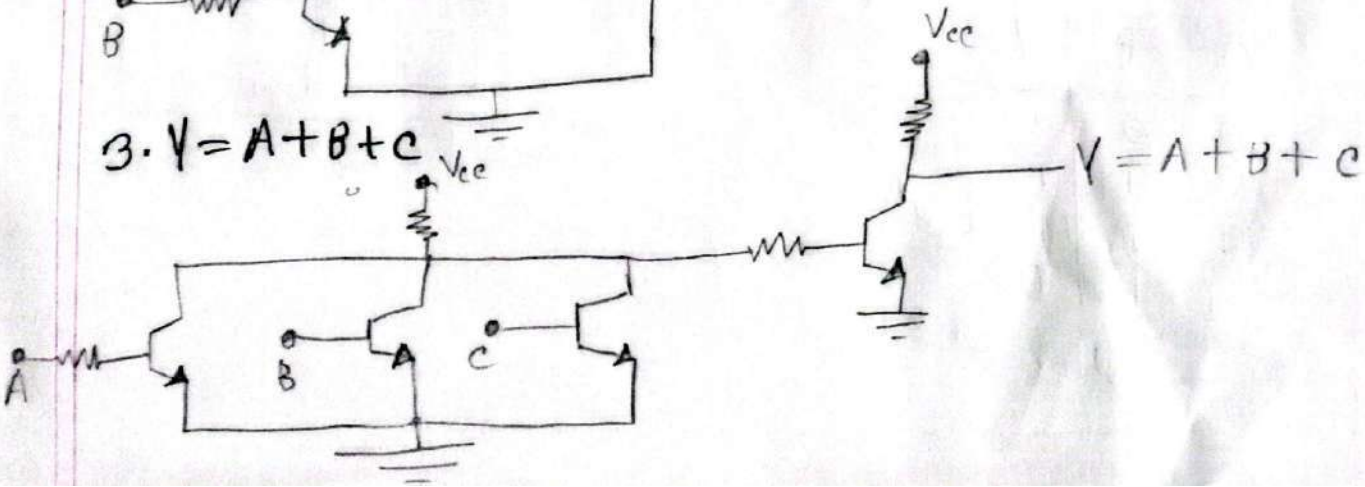
1. $Y = A + BC$



2. $Y = AB + C$



3. $Y = A + B + C$



9 August 2026

O/p - output

DTL

RTL has low noise margin, slow speed & higher Power disipation.

DTL has improved NM compared to RTL

Advantage

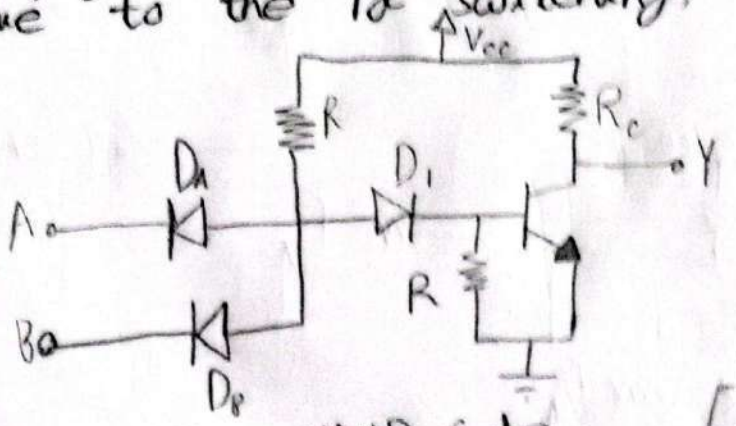
→ O/p Voltage is not reduced when load on gate is increased

→ NM is more

Disadvantage

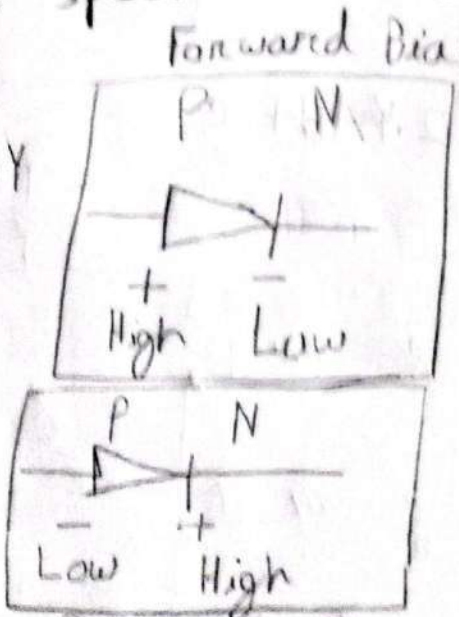
→ Propagation delay is more

→ Due to the Pd switching speed is less

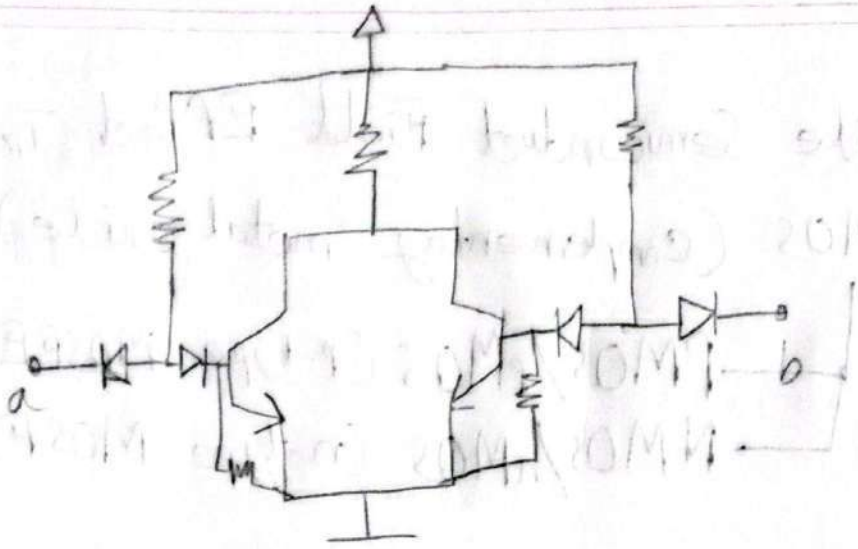


NAND Gate

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



next class: CMOS
PT CT



NOR Gate

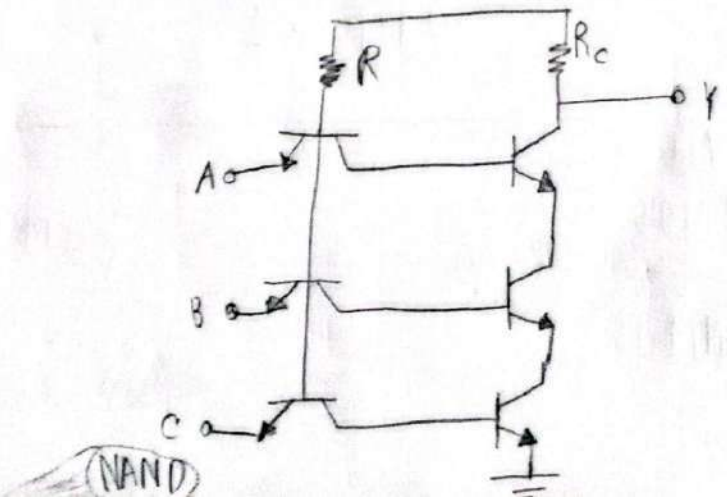
TTL

Advantage

- Circuit gives low o/p impedance
- Gate can drive capacitive load due to low o/p impedence
- power dissipation is less

Disadvantages

Large current spike is produced while switching



NAND Gate			
A	B	C	$\overline{ABC}$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

10 August 2026

MOSFET

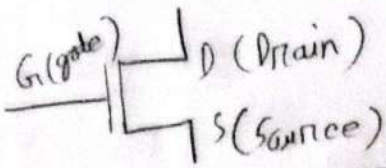
Metal Oxide Semiconduct Field Effect Transistor

└ cMOS (complementry metal oxide)

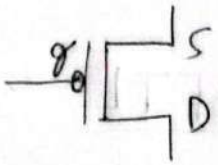
  ├ PMOS/pMOS (p-type MOSFET)

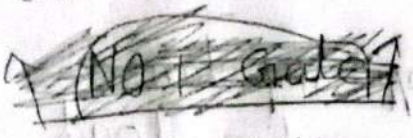
  └ NMOS/nMOS (n-type MOSFET)

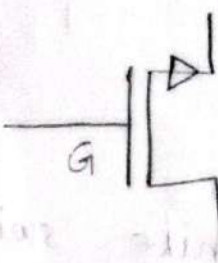
NMOS



PMOS

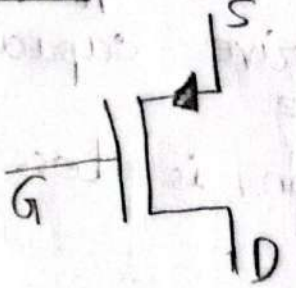






gate → 1

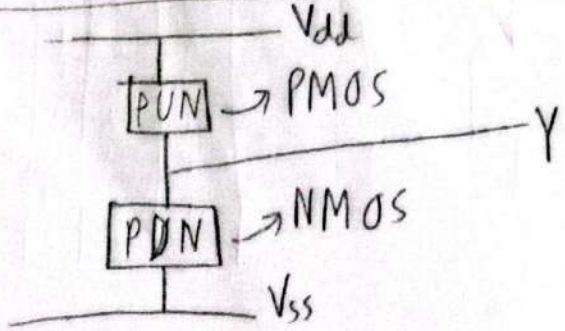
off → 0



hole না থাকলে current flow করবে

ON → 0

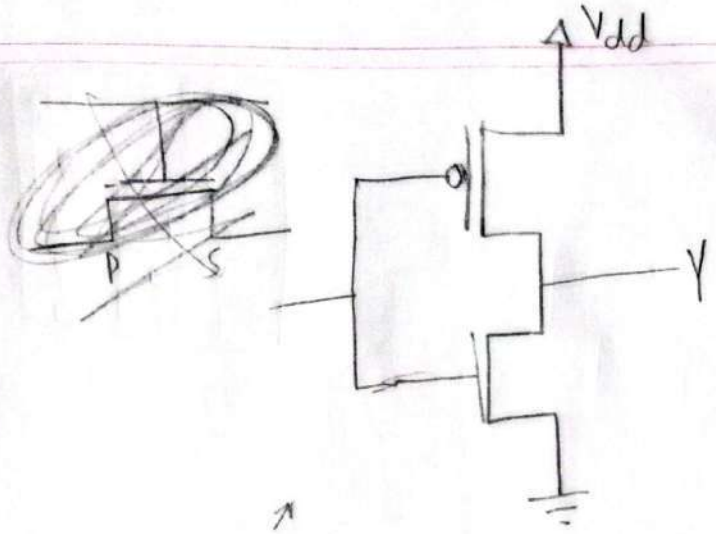
OFF → 1



PUN
Pull up Network

PDN
↓
Down

} cMOS এর structure



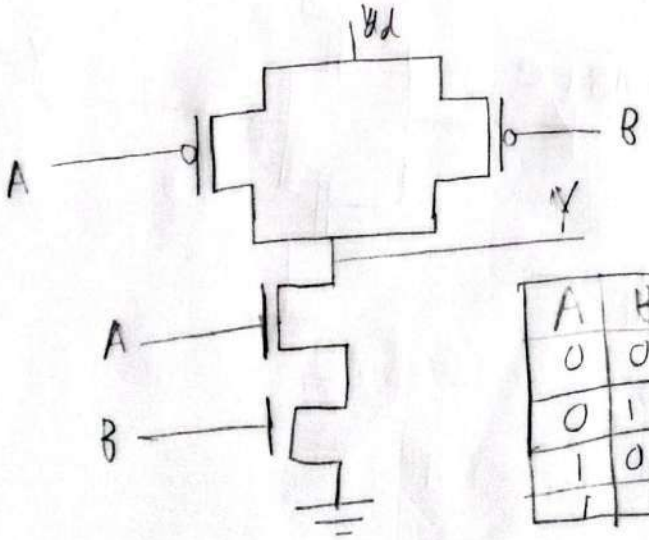
CMOS NOT Gate

2 Input Nand

Rule

Dot(•) হলে PMOS → Parallal
NMOS → Series

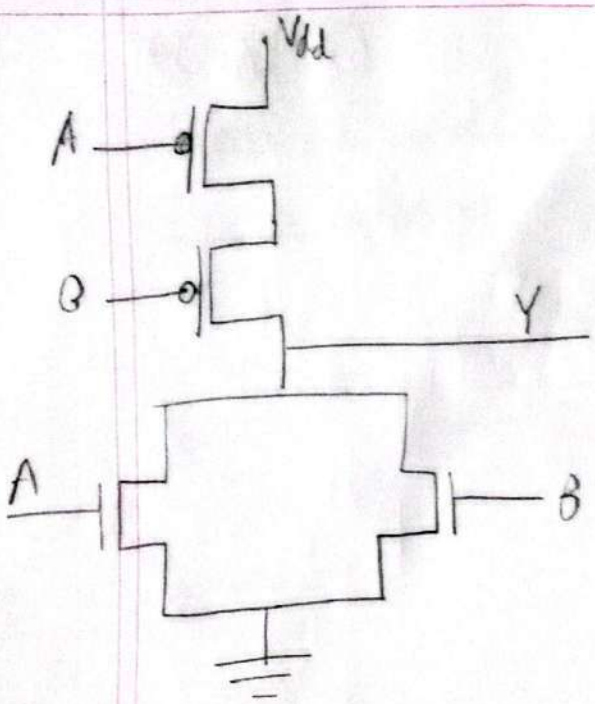
Plus(+) হলে PMOS → Series
NMOS → Parallal



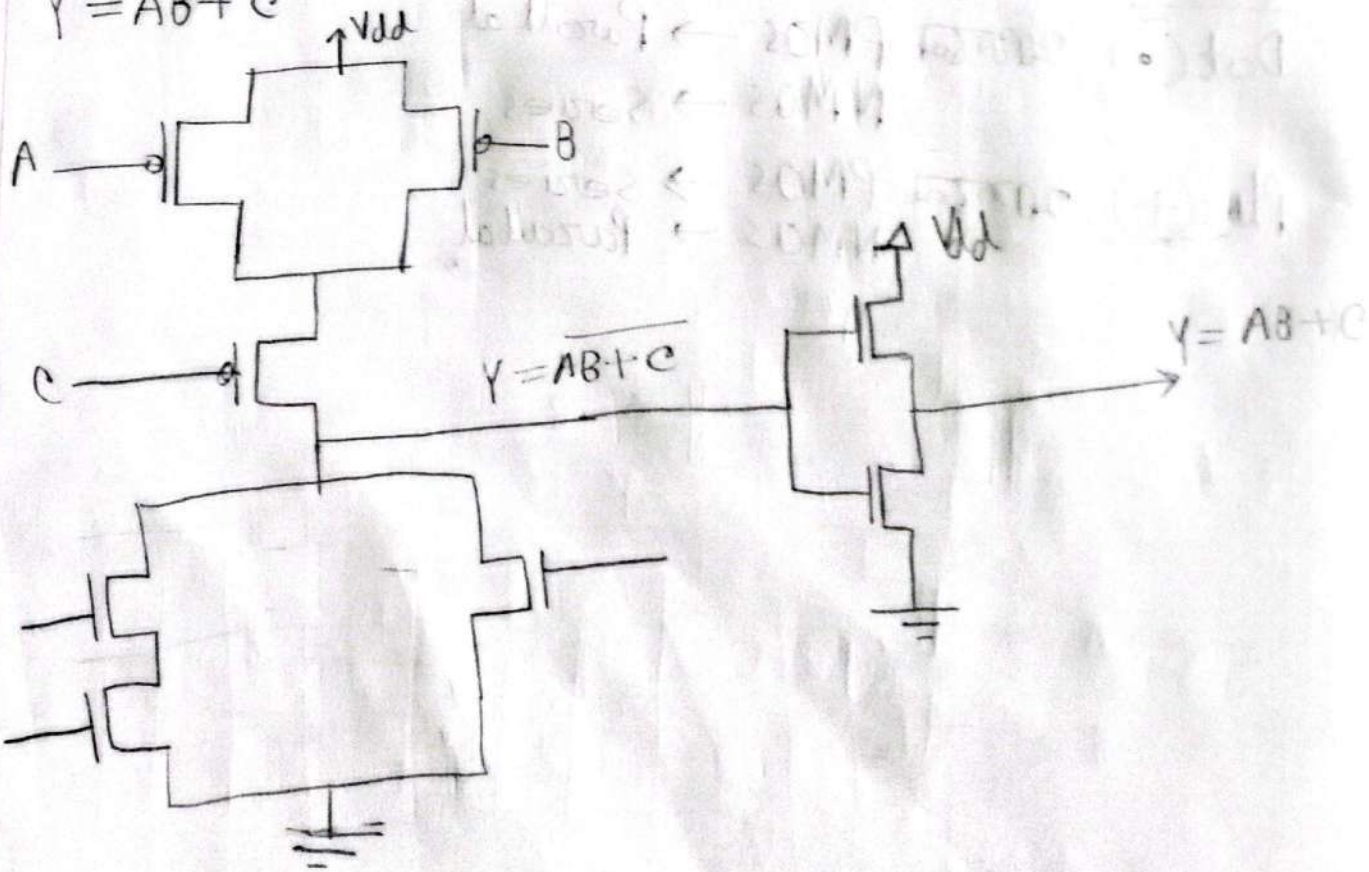
A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

Nand Gate

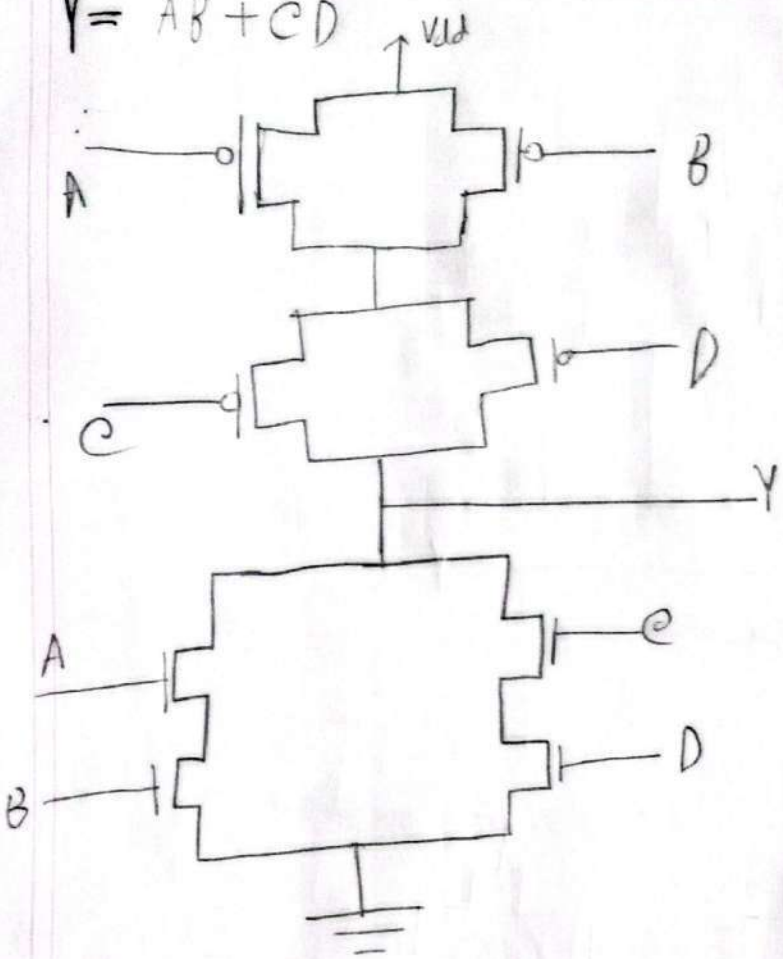
A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0



$Y = AB + C$

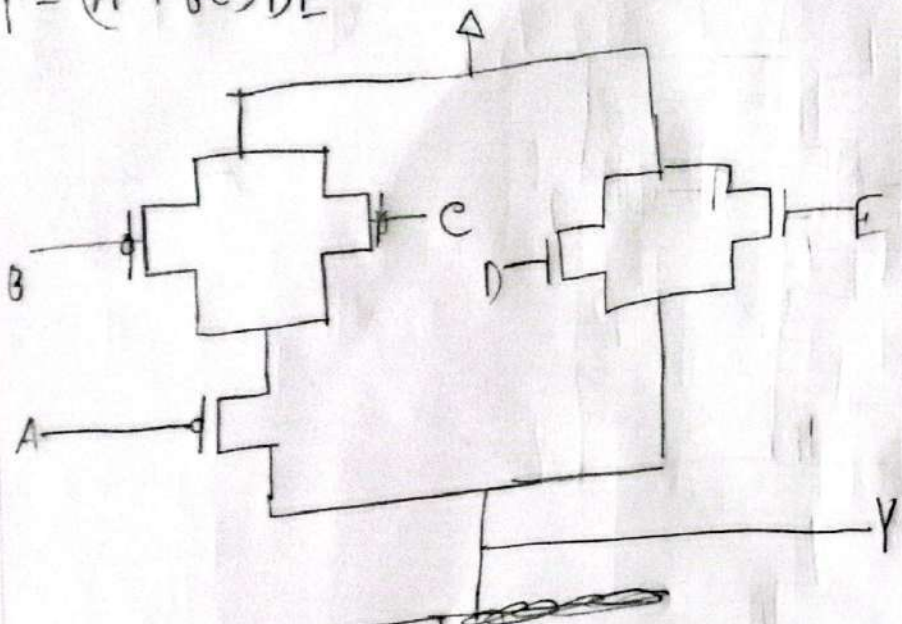


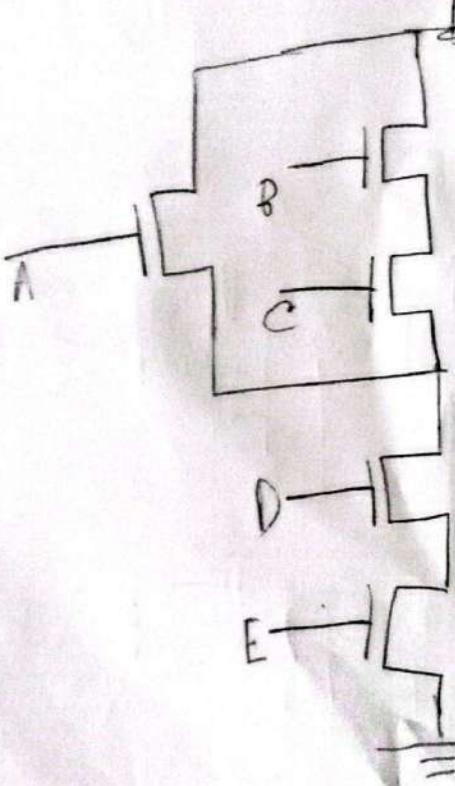
$Y = \overline{AB + CD}$

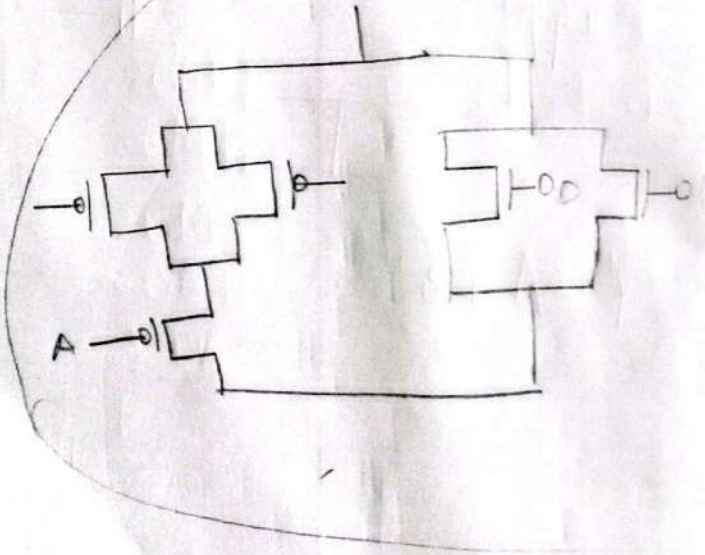


~~Y = A + B~~

$Y = \overline{(A + BC)DE}$



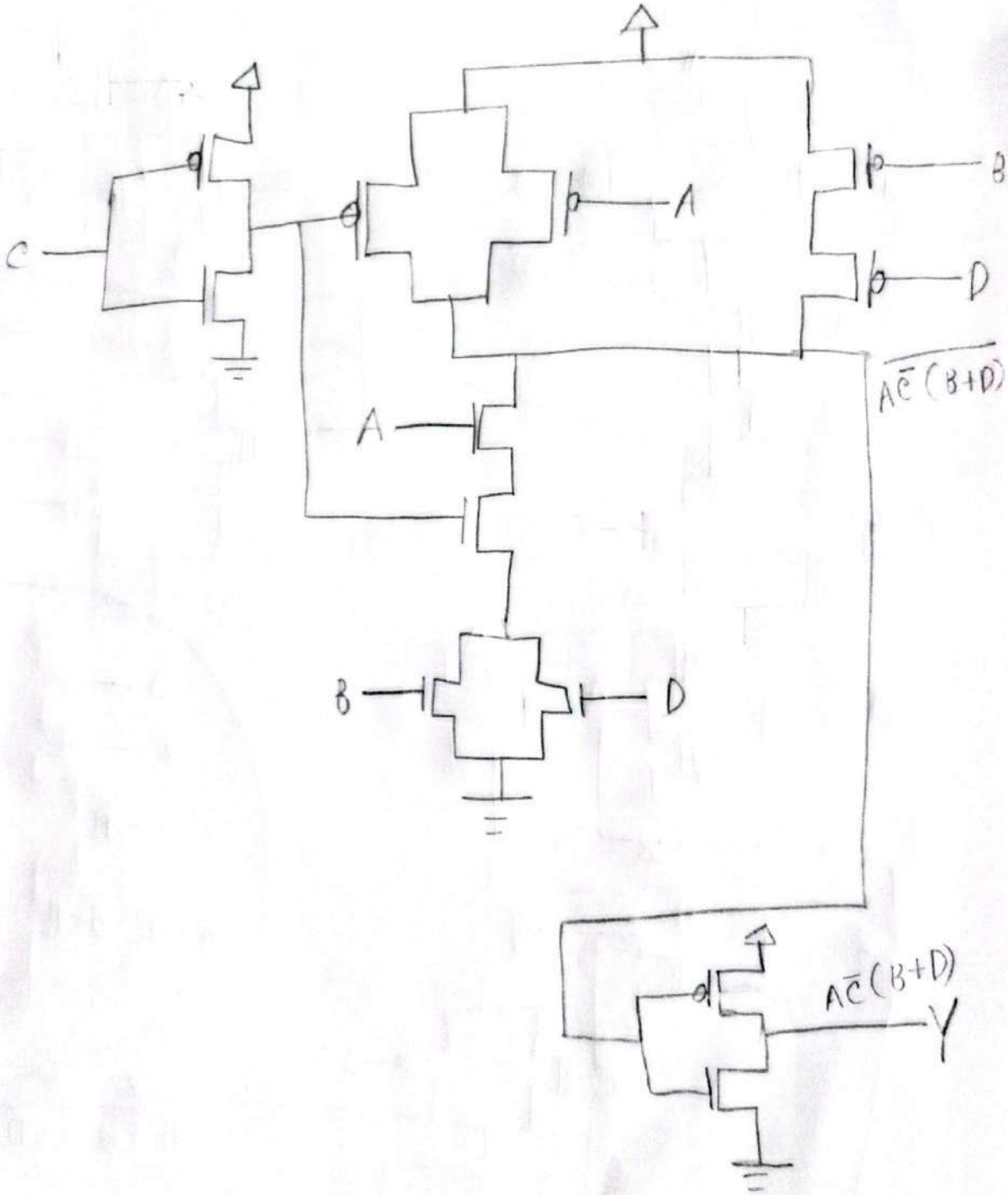




next week এর monday তে CT-1
এগুলন পর্যন্ত যা যা পড়াইছে

$Y = (AB+D)A\bar{C} = AB \cdot A\bar{C} + DA\bar{C}$

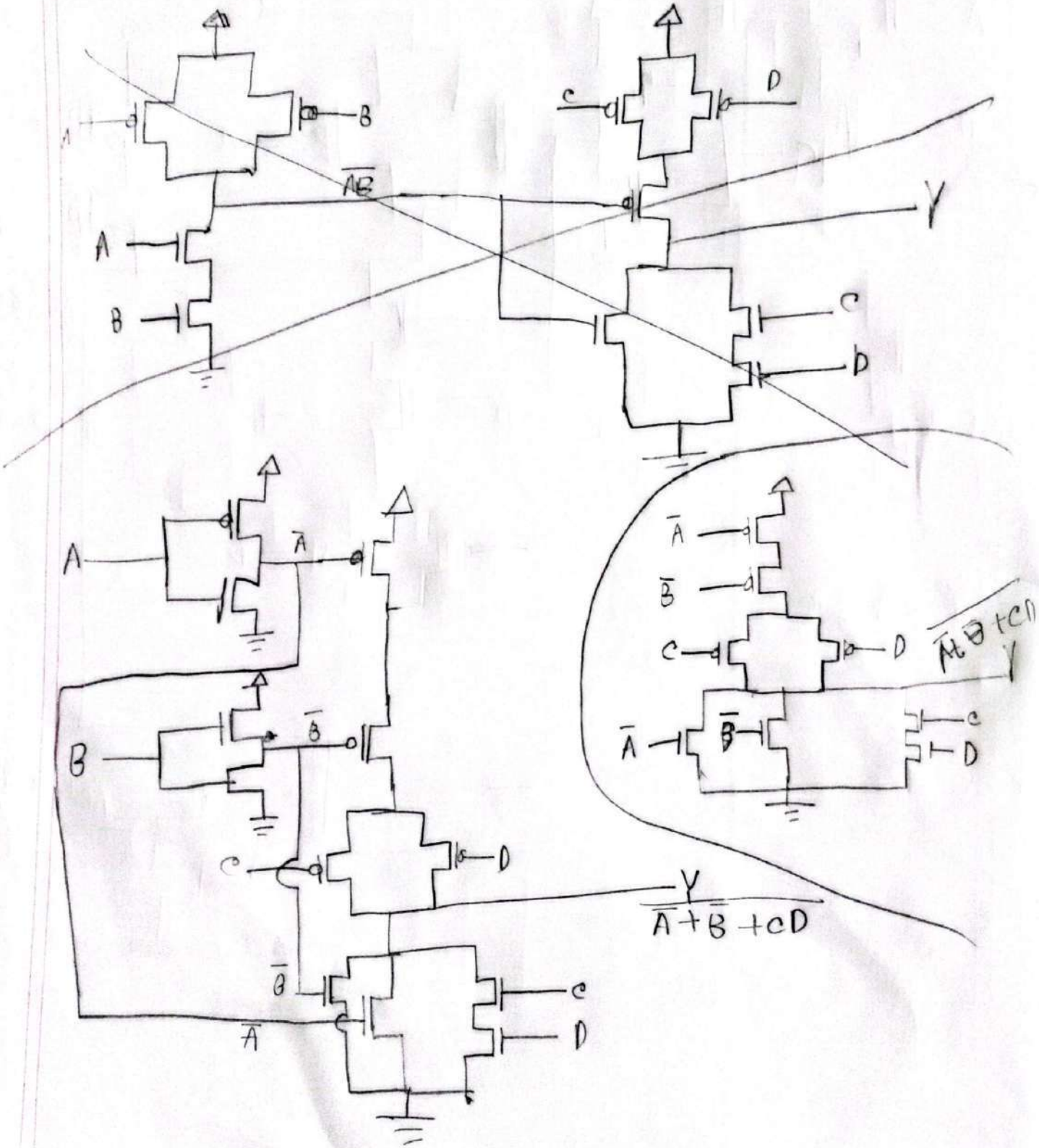
$= AB\bar{C} + AD\bar{C} = A\bar{C}(B+D)$



H.W $F = \overline{D + A(B+C)}$

$F = A\bar{B} + \bar{B}CD$

$Y = \overline{A\bar{B} + CD} = \bar{A} + \bar{B} + CD$



Mid ও electronix ও Machine (50/50)
electronics সম্পূর্ণ compelete (17 August 2024)

18/08/2026

Transformer

Slide ৩ 3

→ A transformer is a static piece of apparatus by which ac electric power of ~~one~~ one voltage level in one circuit is transferred to another circuit at another voltage level maintaining the same frequency

→ Transformer works according to principle of faraday's law of electromagnatic induction

→ B
- ✓ Transfers electric power from one circuit to another.
- ✓ It does so without a change in frequency
- ✓ It accomplishes this

লোয়ার আগের Slide changed

Slide ৪ 4
1885 (পুরনো Slide এ 1985 ছিলো)
Transfer invented

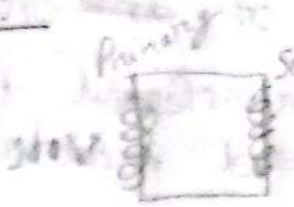
Slide ৫ 5
Variation এগুলোই Transformer এর

Slide ৬ 6
DC power was mainly is used in the 1880's and it was hard to transmit over distance
DC তে tranformer আগে ব্যবহৃত হতো না। সুবিধা ছিল।
এখন AC use হয়।

Slide 6: (কিছু বাদ দিও / কত ... এর না একদম বড়ো করে বোঝাও আছে)

Slide 7: [illegible] সার্কিট

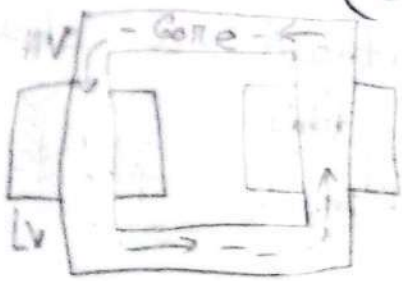
Slide 8:



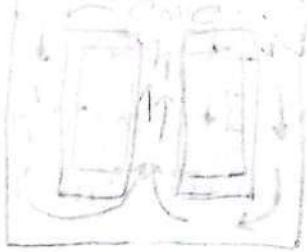
Transformer এর input side fixed না।
(Primary, Secondary) = (input, output)

Tranforme দুই type

1. Core-type transformer (দুই দা L)
2. Shell-type transformer (E এর মাঝে I)



Core-type Construction



Shell-type construction

Faraday's law of electromagnatic induction

Magnatic flux ↑ magnatism	electron ↑ electricity

Slide 10

Lickege flux

Slide 11

এই সব ব্যাপার দরকার নাই।

Slide 16

E, I এর চিত্র পাবেন Slide এ

24/8/2026

L 2 - Transformer.pdf

Slide 22

Ideal Transformer (Loss নাই) (100% efficient)

~~Static (কোন Mechanical sector নাই)~~

Slide 23

i.

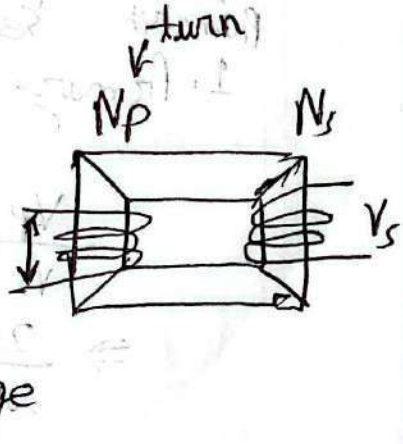
(ii)

(iii) Core-loss zero

(iv) core has infinite permeability.

(Magnetic flux flow হবে কিন্তু কম বেশি)

Slide 24 P - Primary, S - secondary



$$\frac{V_p(t)}{V_s(t)} = \frac{N_p}{N_s} = a = \frac{I_s}{I_p}$$

$$N_p i_p(t) = N_s i_s(t)$$

Slide 25

$$\frac{V_p}{V_s} = a \qquad \frac{I_p}{I_s} = \frac{1}{a}$$

~~Slide 28~~

Slide 28

$\theta_p = V_p - I_p$ (angle) $P = VI\cos(\theta_V - \theta_I)$

$$P_{in} = V_p I_p \cos\theta_p$$

$$P_{out} = V_s I_s \cos\theta_s$$

$$P_{out} = \eta \cdot P_{In}$$

Textbook of electrical Technology

~~Text~~ Book Slide ~~No~~ 267 — 32.1

The maximum flux density in the core of a 250/3000 volts, ~~50 Hz single-pha~~ E.M.F per turn is 8 volt,

(Primary, Secondary)

1. Primary and Secondary turns =

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\Rightarrow \frac{250}{3000} = \frac{N_p}{N_s}$$

$$\Rightarrow \frac{1}{12} = \frac{N_p}{N_s}$$

$$\frac{I_s}{I_p} = \frac{V_p}{V_s} = \frac{1}{12}$$

$$\Rightarrow \frac{I_p}{I_s} = 12$$

Slide 32

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

$K > 1 \rightarrow$ Step-up transformer

$K < 1 \rightarrow$ step-~~down~~ down transformer

$K = 1 \rightarrow$ Ideal transformer

Book. 267 – 32.1 (i)

$E_1 = N_1 \times$ e.m.f induced/turn

$$\Rightarrow N_1 = \frac{E_1}{\text{e.m.f induce/turn}}$$

turns সর্বদা ১ না

$$\Rightarrow N_1 = \frac{250}{8} = \cancel{31.25} = 32$$

$$N_2 = \frac{3000}{8} = 375$$

32.2 Voltage 11000/500V ; Emf 10.4 ← প্রতি পিস

$$N_1 = \frac{11000}{10.4} = 1058$$

$$N_2 = \frac{500}{10.4} = 49$$

$$\frac{I_1}{I_2} = \frac{V_2}{V_1} = \frac{500}{11000} = \frac{1}{20}$$

$$\cancel{\frac{I_1}{I_2} = \frac{N_2}{N_1} = \frac{49}{1058} = 0.046}$$

$$\Rightarrow \frac{I_2}{I_1} = \frac{1058}{49} = 2$$

15/5/2026

Book Page 272

1. The number of turns on the primary and secondray of a 1−ϕ tranformer are 350 and 35 respectively. If primary is connected to a 2.2kV, 50Hz suply, determine the secondry voltage on load.

We know,

$$\frac{N_1}{N_2} = \frac{V_1}{V_2}$$

$$\Rightarrow V_2 = \frac{N_2}{N_1} \times V_1$$

$$= \frac{35}{350} \times 2200$$

$$= 220\,V$$

Primary $I = 5A$

Secondry $= I_2$

$$\frac{I_2}{I_{o1}} = \frac{V_1}{V_2} = \frac{2200}{220} = 10$$

$$\Rightarrow I_2 = \frac{1}{10} \times I_1$$

$$I_2 = \frac{1}{10} \times 5$$

$$= 50\,A$$

4(a) A 40 kVA, 3,300/240V, 50 Hz, 1 phase transformer has 660 turns on the primary.

⇒ Determine the number of turns on secondary.

$$\frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{I_s}{I_p}$$

Motor Inductive Load

Fan

We know, $\frac{N_1}{N_2} = \frac{V_1}{V_2}$

$$\Rightarrow \frac{660}{N_2} = \frac{3300}{240}$$

$$\Rightarrow N_2 = \frac{240}{3300} \times 660$$

$$= 48$$

2(b) A 3000/200 V, 50 Hz, 1 phase transformer ..... area of 150 cm^2 and has 80 turns in the low-voltage welding. Calculate num turn in high-voltag

We know,

$$\frac{N_1}{N_2} = \frac{V_1}{V_2}$$

$$\Rightarrow \frac{N_1}{80} = \frac{3000}{200}$$

$$\Rightarrow N_1 = \frac{3000}{200} \times 80 = 1200$$

Active power
Reactive power
Apparent power

Volt Ampere Reactive

Volt Ampere

$P = VI\cos\theta$ W

1 KWh

100 W X 10 h

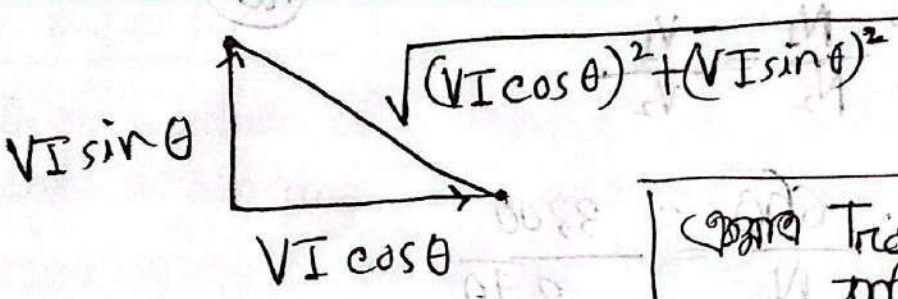
$P = VI \sin\theta$ VAR

$P = VI$ VA

Power ত্রিভুজ

ফিল্টার - Real (Active)
বাতাস - Reactive

Reactive মাধ্যমে Real এর flow করার জন্য আমাদের করতে



এজন্য Transformer VA তে রেটিং করা হয় Watt এ নয়

Why tranformers are rated in VA not in Watt

7 september CT-2

31/8/2026

X-reactance

Slide 40

(I) Core or iron loss

Slide 41

Copper Loss (Heating loss I^2R)

Slide 58

Equivalent circuit of transformer

Resistor ও resistance বলতে

Inductor " " বলতে Inductance ও বোঝায়।

Ω

H

Henry to ohm

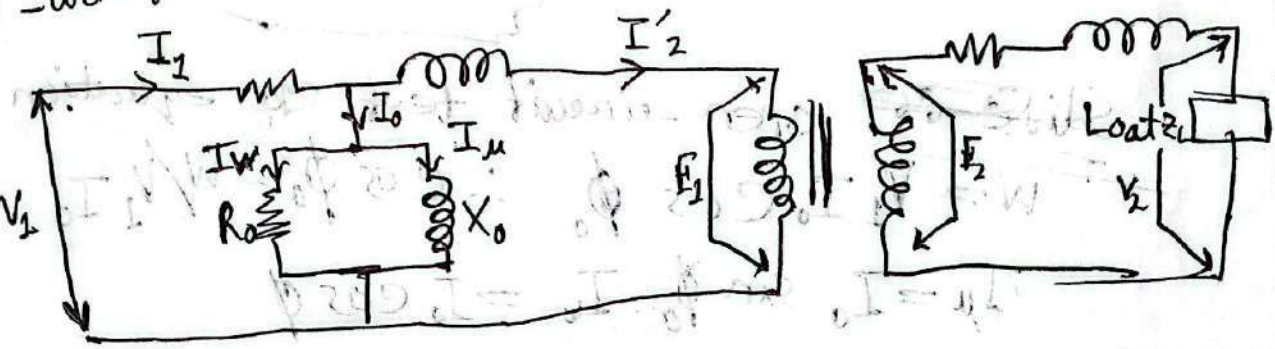
$2\pi fL = \omega L$

$= X_L$

Ω

Primary – secondary

I_w - working



I_0 - No load current

I_w - working component

I_μ - Magnetizing component

variac = variable ac

wattmeter

Slide 59 → simplified
Slide 60 → simplied
} এই তিনটি circuit নিয়ে আলোচনা

Slide 61

Slide 63

Transformer এর rating VA তে প্রকাশ করা হয়। কেন প্রকাশ করা হয়?

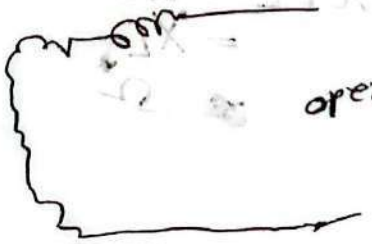
Slide 64

open - circuit test → High তে supply দেওয়া হয়
short " → Low voltage side short করা হয়

Slide 65/66

open - circuit test → High voltage side open থাকে, Low তে supply





Slide 66

open - circuit test এর equation

$W = V_1 I_0 \cos\phi_0 \quad \therefore \cos\phi_0 = W/V_1 I_0$

$I_\mu = I_0 \sin\phi_0 \quad I_w = I_0 \cos\phi$

$\therefore X_0 = V_1/I_\mu$ and $R_0 = V_1/I_w$

আজকে পর্যন্ত CT ক্লিয়ার করার Motor ও Transformer related Basic que

Open circuit test

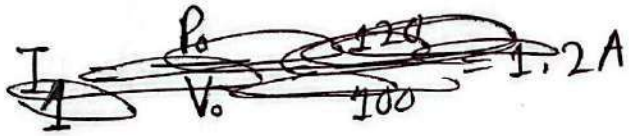
$V_o = 100\ V$

$I_o = 2\ A$

$P_o = 120W$

$R_o = ?$

$X_o = ?$





$$\cos\phi_o = \frac{P_o}{V_o I_o} = \frac{120}{100 \times 2} = 0.6 \quad \therefore \phi_o = 53.13$$

$$I_u = I_o \sin\phi_o = 2 \times \sin(53.13) = 1.6$$

$$I_w = I_o \cos\phi = 2 \times 0.6 = 1.2$$

$$X_o = \frac{V_o}{I_u} = \frac{100}{1.6} = 62.5$$

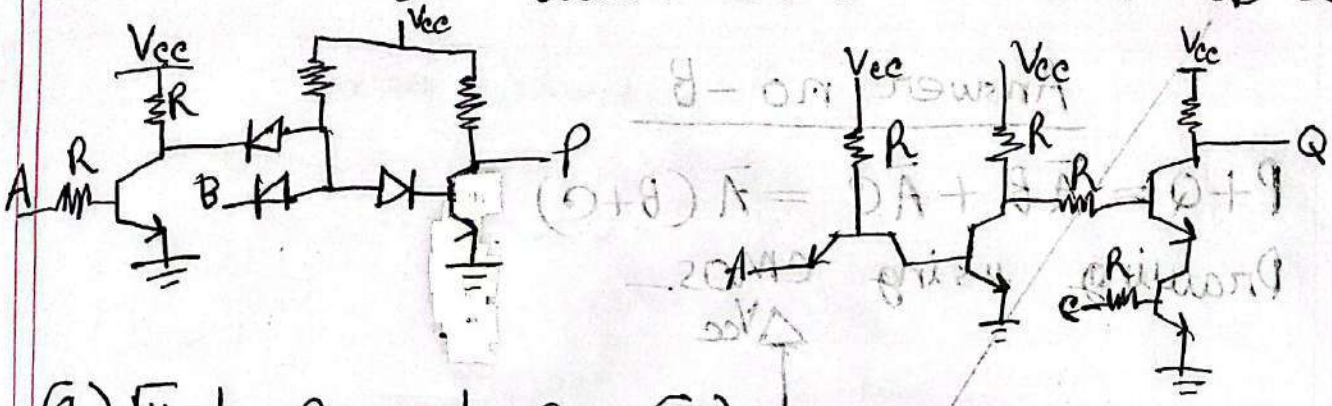
$$R_o = \frac{V_o}{I_w} = \frac{100}{1.2} = 83.33$$

CamScanner

56 Batch sec C CT-1

1. In active mode, the base-emitter junction is forward biased.
2. The current gain of BJT is represented by the symbol β.
3. For amplication, the BJT operates in the active region.
4. BJT has three terminals : Emitter, Base and Collector.
5. The relation between current in BJT is $I_E = I_C + I_B$.
6. The input junction of BJT, is Base-Emitter
7. RTL has relatively low noise margin compared to modern logic families.
8. RTL circuits use resistors and transistors as components.
9. Compared to RTL, DTL has higher noise margin
10. DTL does not use transistor. (True/False)
    Ans: False

56 Batch CT-1 section (D) set (B)



(a) Find P and Q (5)

(b) Draw a circuit for P+Q using CMOS (15)

Answer to the question no-a

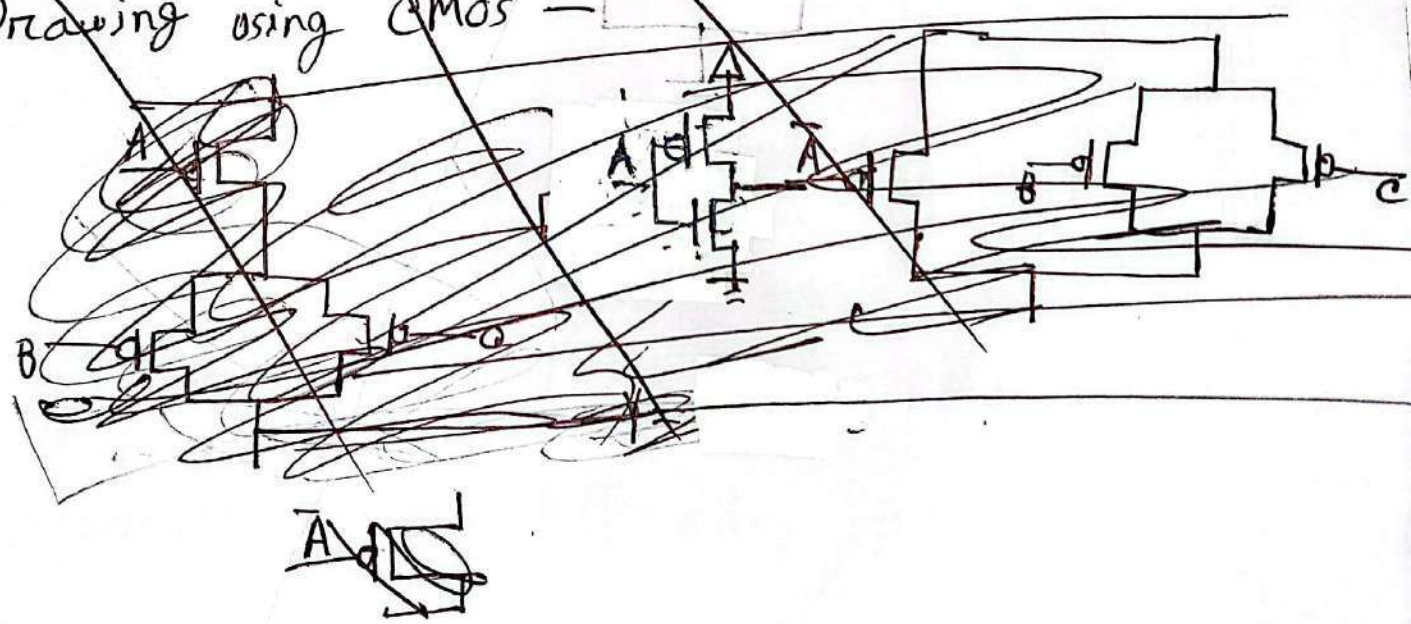
$P = \bar{A}B$

$Q = \bar{A}C$

Answer no-b

$P+Q = \bar{A}B + \bar{A}C = \bar{A}(B+C)$

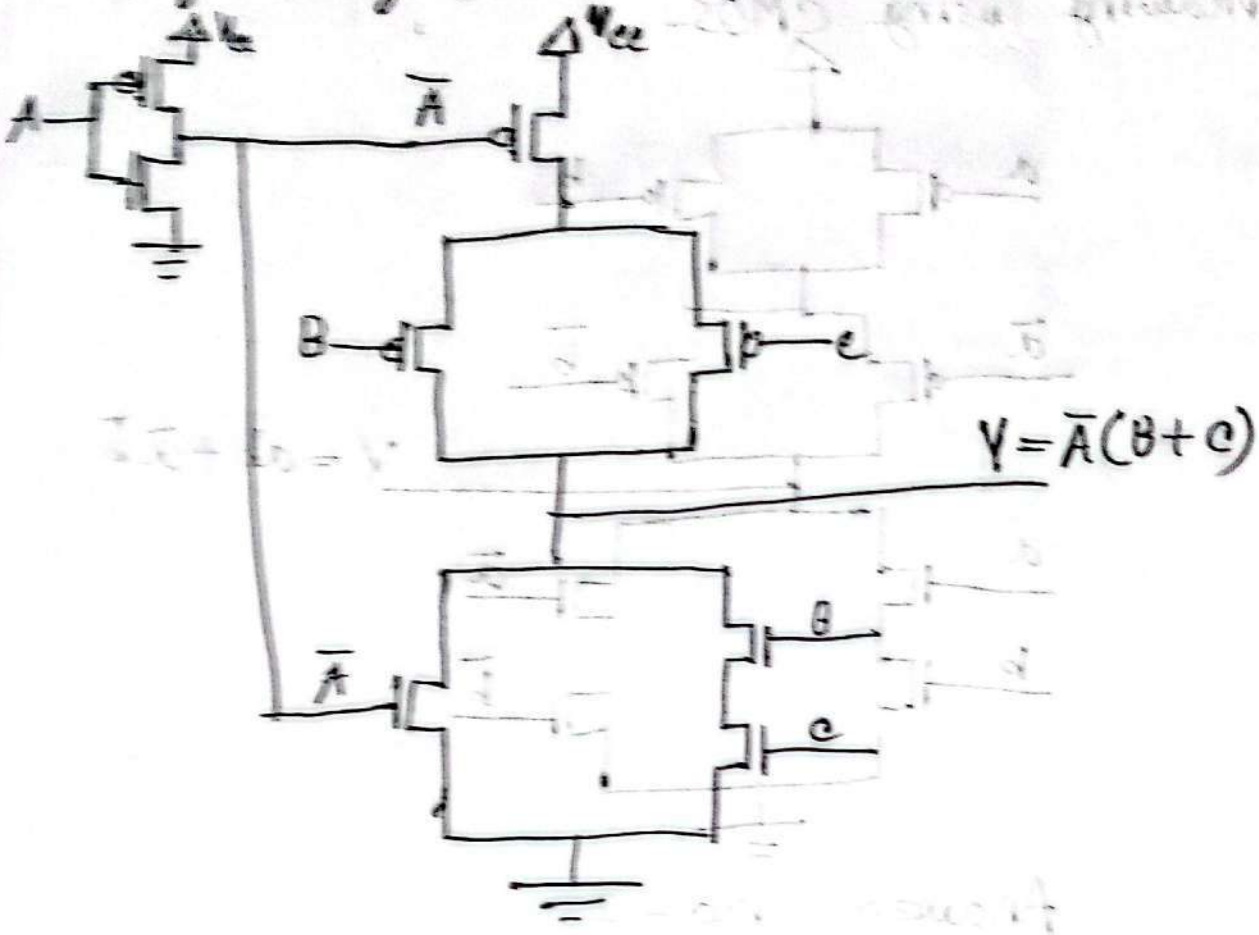
Drawing using CMOS —



Answer no-8

$P+Q = \bar{A}B + \bar{A}C = \bar{A}(B+C)$

Drawing using CMOS-



56th CT-1 section D ~~set A~~

Question 1;

Implement an XNOR gate using CMOS

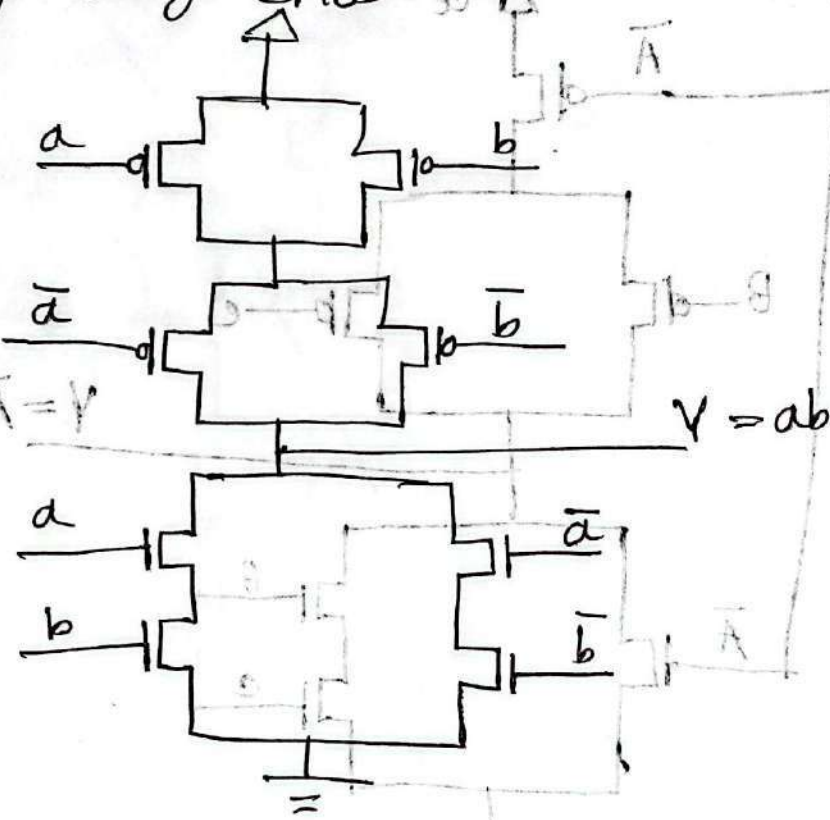
Question 2:

Implement $Y = A + B\bar{C}$ using DTL

Answer to the question no-1

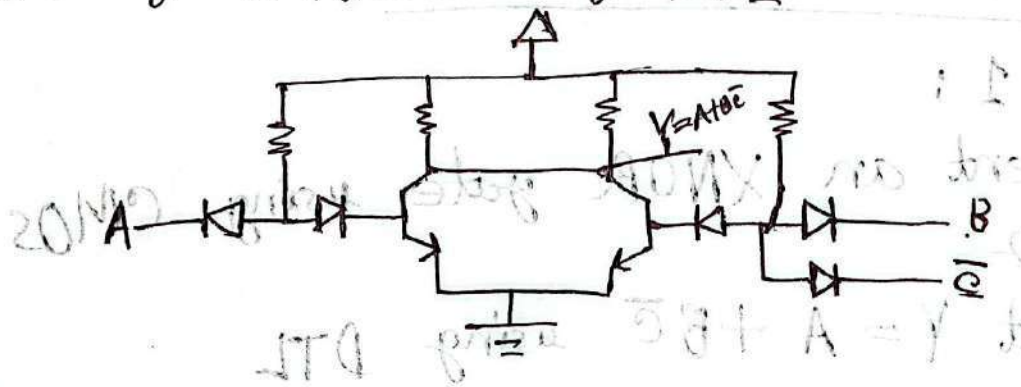
For XNOR gate $Y = ab + \bar{a}\bar{b}$

Drawing using CMOS-



Answer no - 2:

Drawing $A + B\bar{C}$ using DTL



55 Batch section D

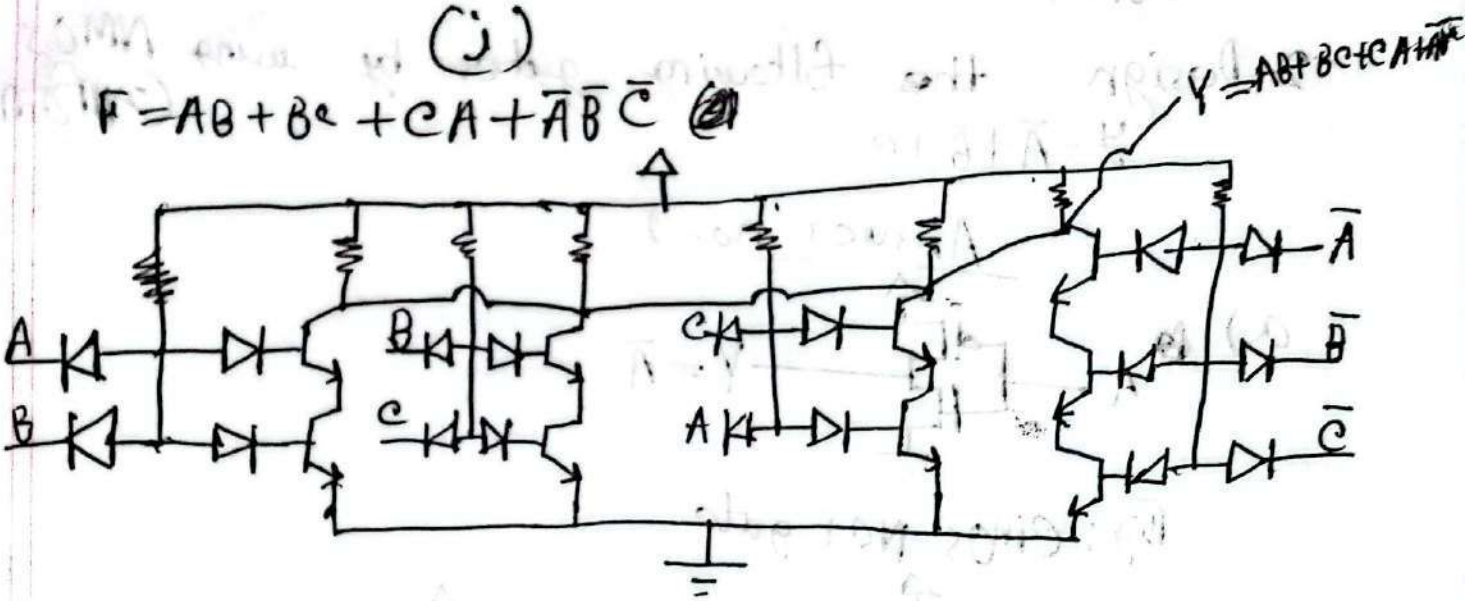
1. Design the Basic Gates using

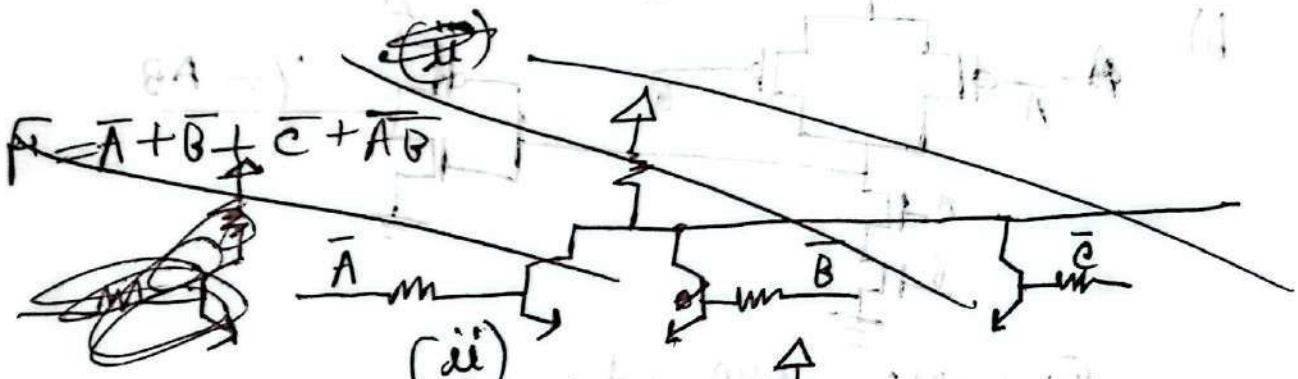
(i) $F = AB + BC + CA + \bar{A}\bar{B}\bar{C}$ (DTL)

(ii) $F = \bar{A} + \bar{B} + \bar{C} + \overline{AB}$ (RTL)

(i)

$F = AB + BC + CA + \bar{A}\bar{B}\bar{C}$





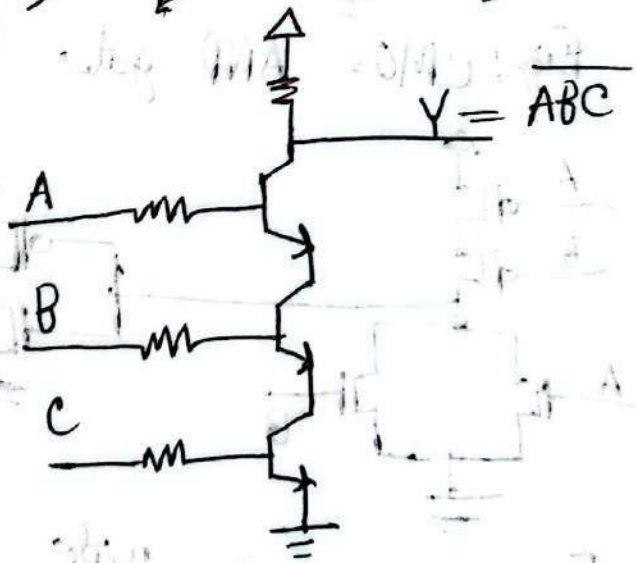
(ii)

$F = \bar{A} + \bar{B} + \bar{C} + \overline{AB}$

$= \bar{A} + \bar{B} + \bar{C} + \bar{A} + \bar{B}$

$= \bar{A} + \bar{B} + \bar{C}$

$= \overline{ABC}$



54 Batch section-C

1. Design the following gates by using CMOS
   a) NOT gate
   b) AND gate
   c) OR gate
   d) XOR gate

2. Design the following gates by using NMOS (any)
   $Y = \bar{A} + \bar{B} + \bar{C}$

Answer no-1

a)

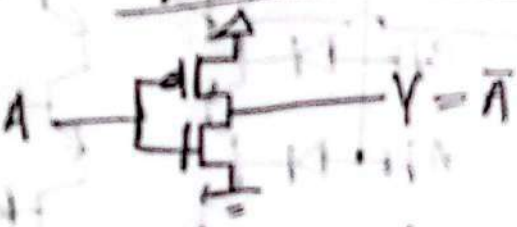


Fig: CMOS NOT gate

b)

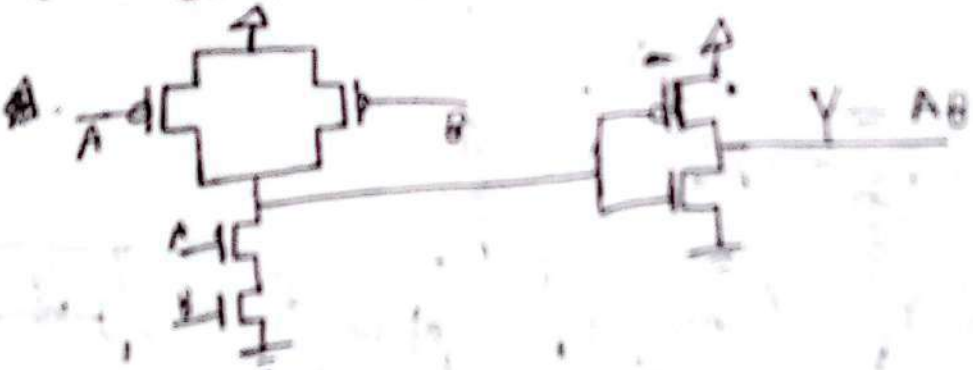


Fig: CMOS AND gate

c)

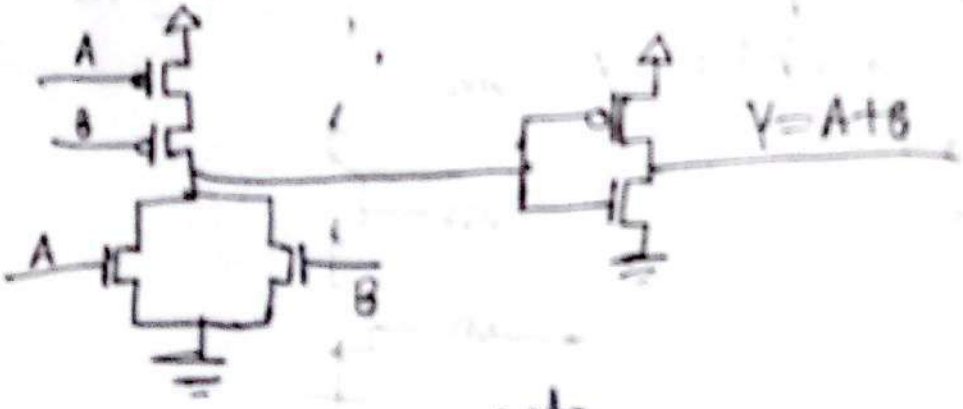
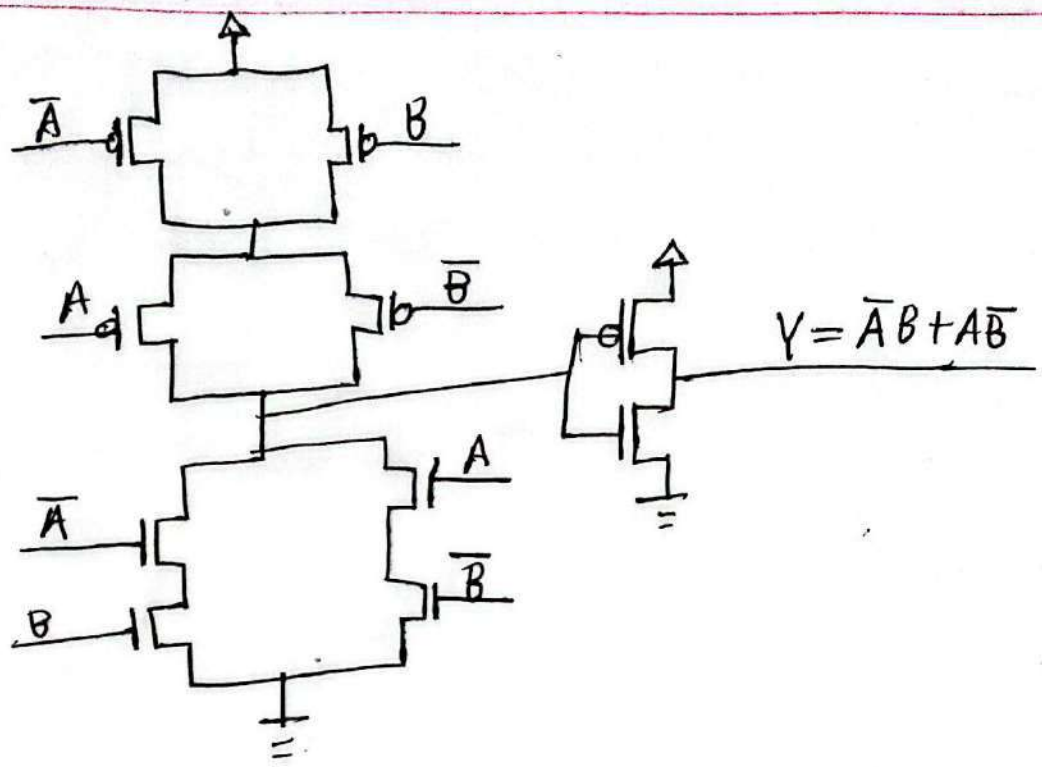


Fig: CMOS OR gate

d)



53 Batch section c

1. Desine the basic circuits by RTL, TTL and CMOS
2. Desigh the circuit $Y = ABC$

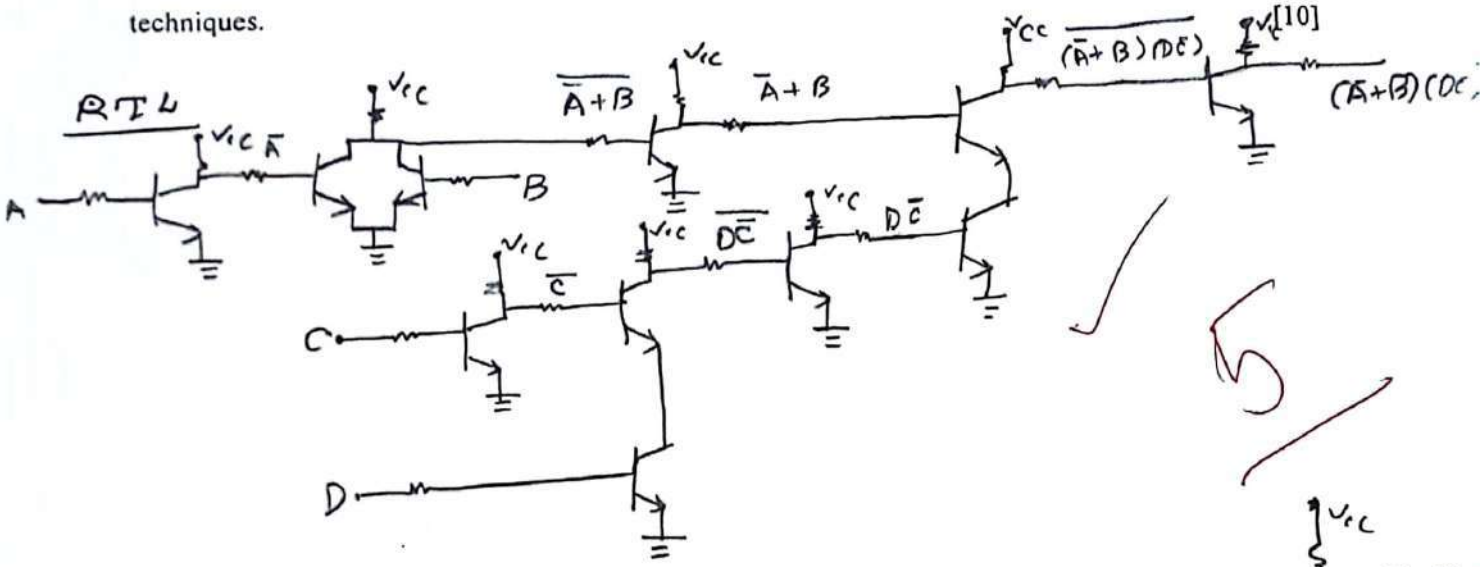
University of Asia Pacific
Department of Computer Science & Engineering
Class Quiz 1, Fall-2023

15

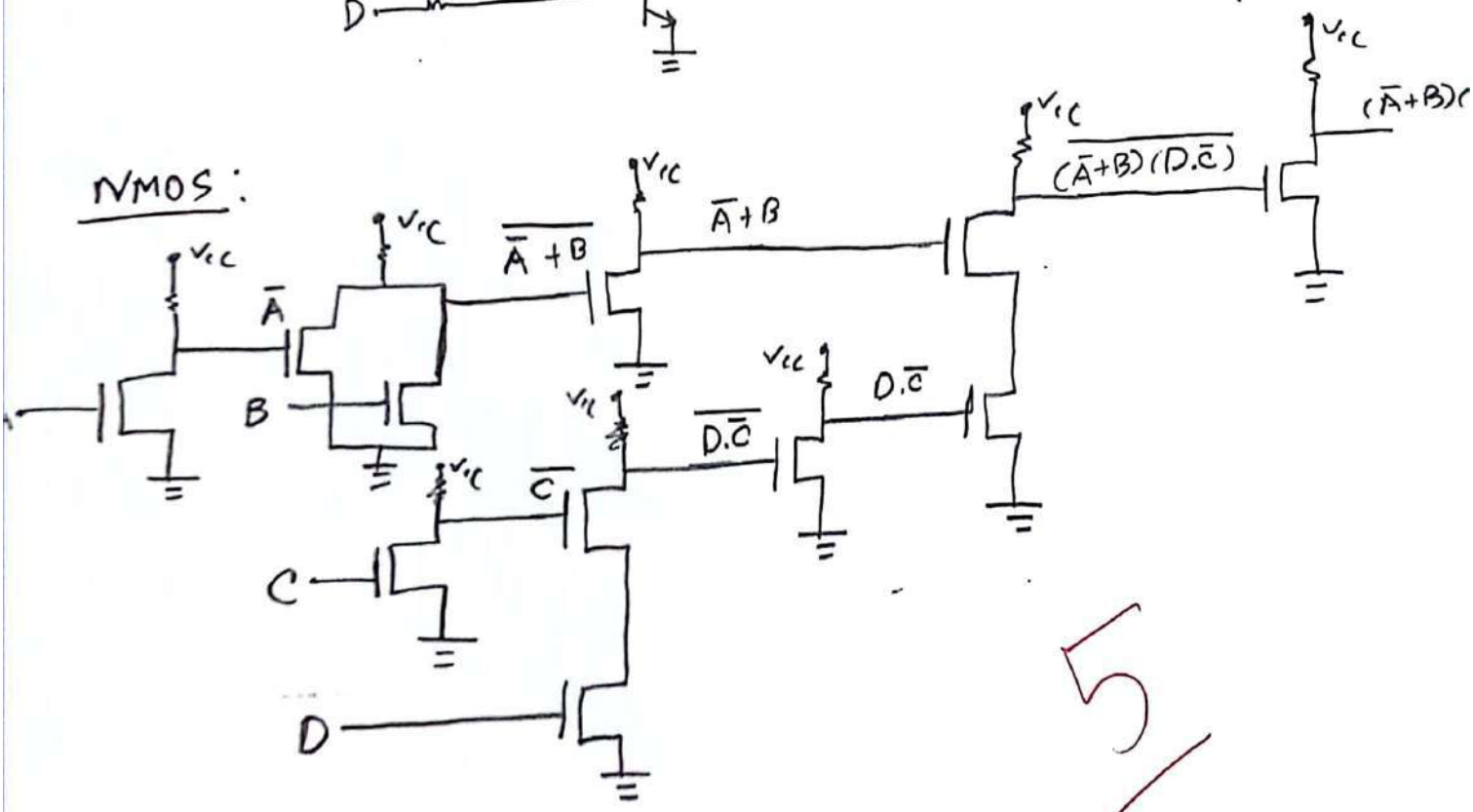
Course Title: Electronic Devices & Circuits | **Course Code: EEE 201**
Time: 30 min | **Full Marks: 15**

Name: MD. Sydul Islam Zubair | ID: 22201174

1. Construct the circuit for the logic function, $F = (\bar{A} + B)(D\bar{C})$ using RTL and NMOS design techniques. [10]



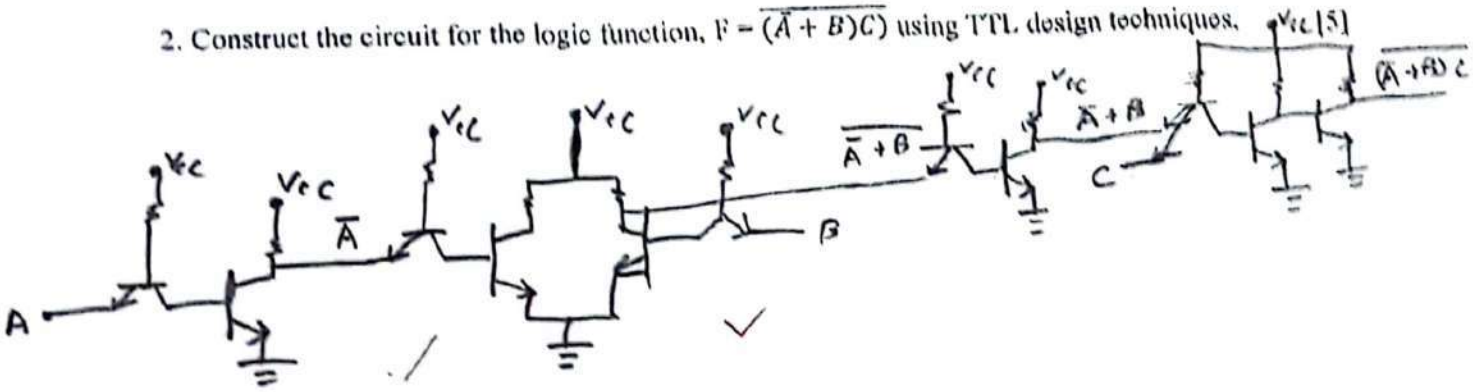
5



5

CamScanner

2. Construct the circuit for the logic function, $F = \overline{(\bar{A} + B)C}$ using TTL design techniques. [5]



5

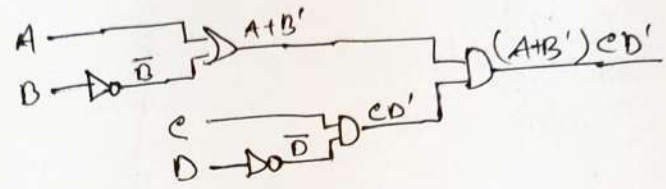
Md. Junaid Hossain
22201017

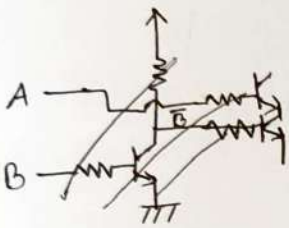
Quiz 1
EEE 201
Set B

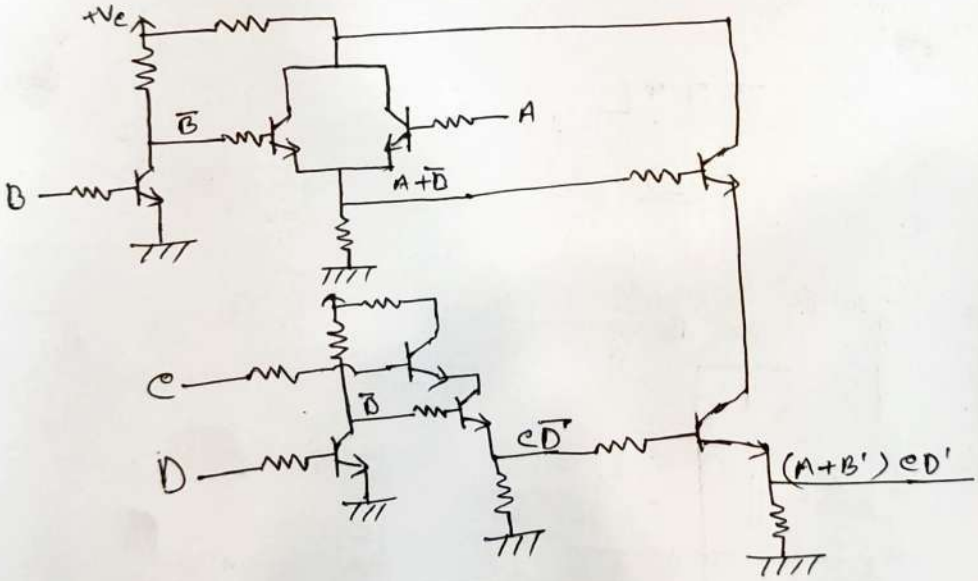
1. Design logic circuit using TTL and RTL for the expression Y= (A+B')D'C
2. Design logic circuit using CMOS for Y= A'B+CD

(1)

$(A+B')D'C$

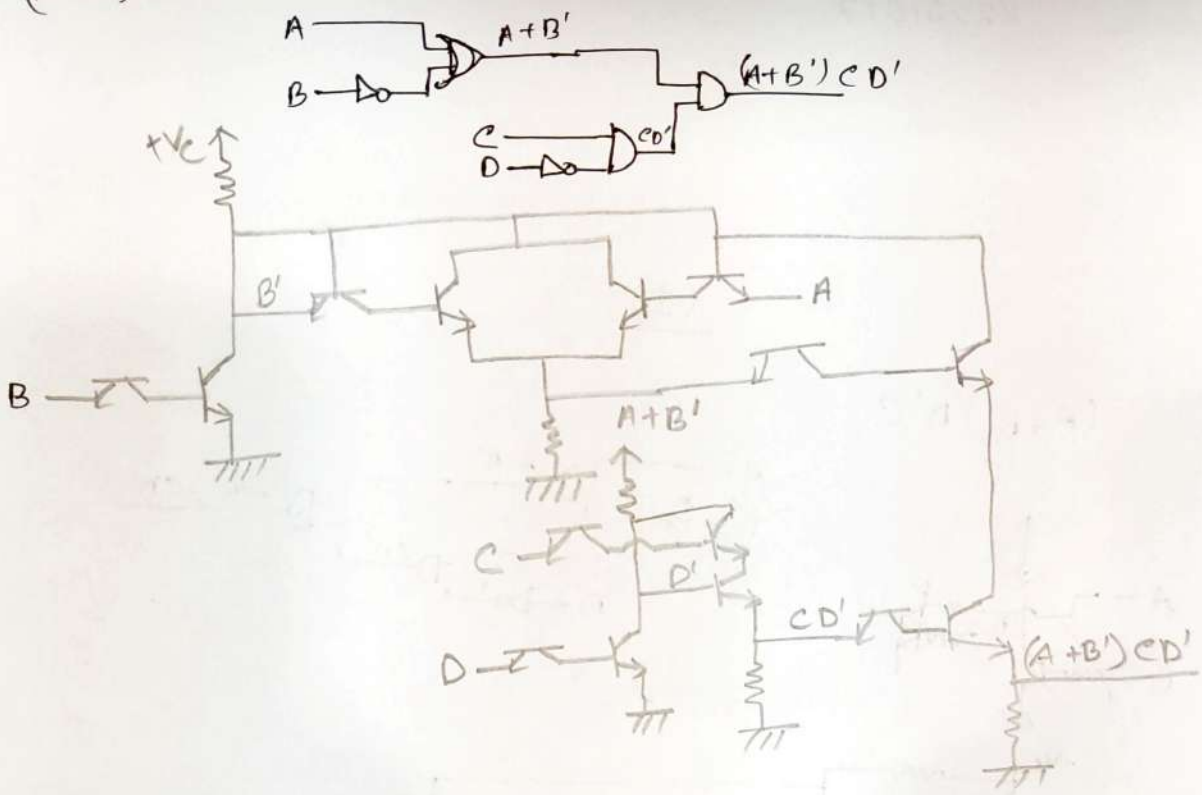






RTL

$(A+B')\,CD'$



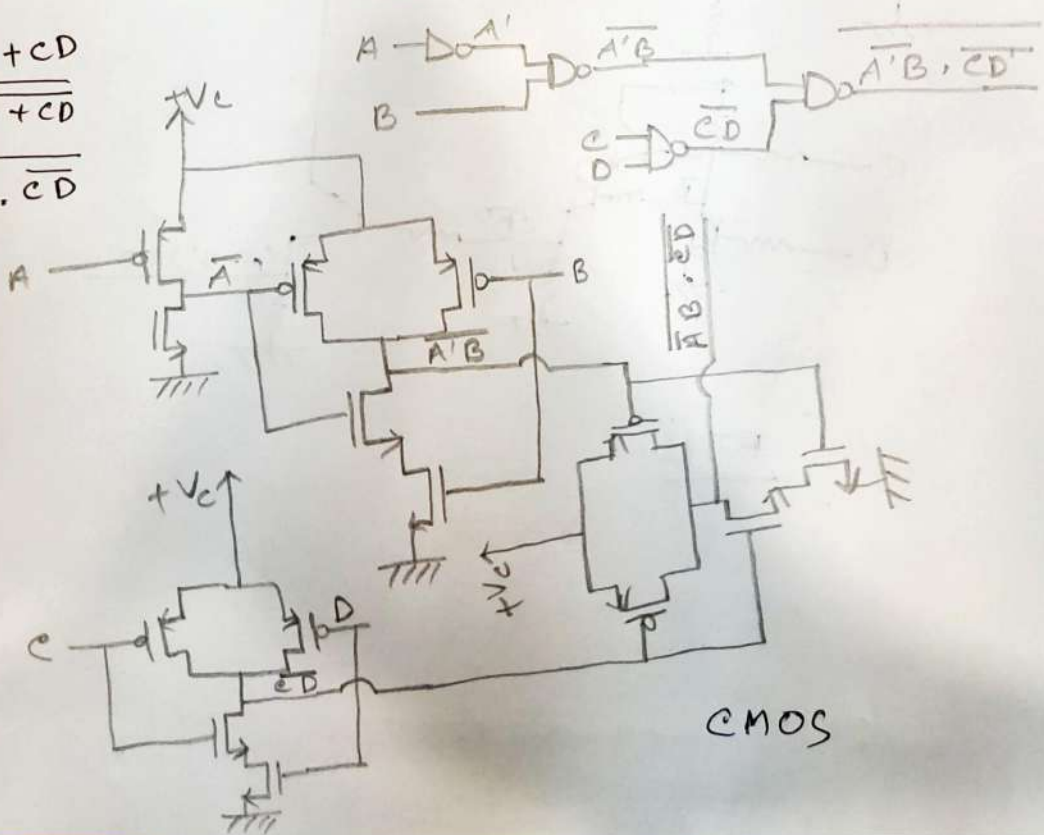
TTL

(2)

$A'B+CD$

$= \overline{\overline{A'B+CD}}$

$= \overline{\overline{A'B}\cdot\overline{CD}}$



CMOS

Quiz 1
EEE 201
Set A

Marzan Ahmed.
22201055

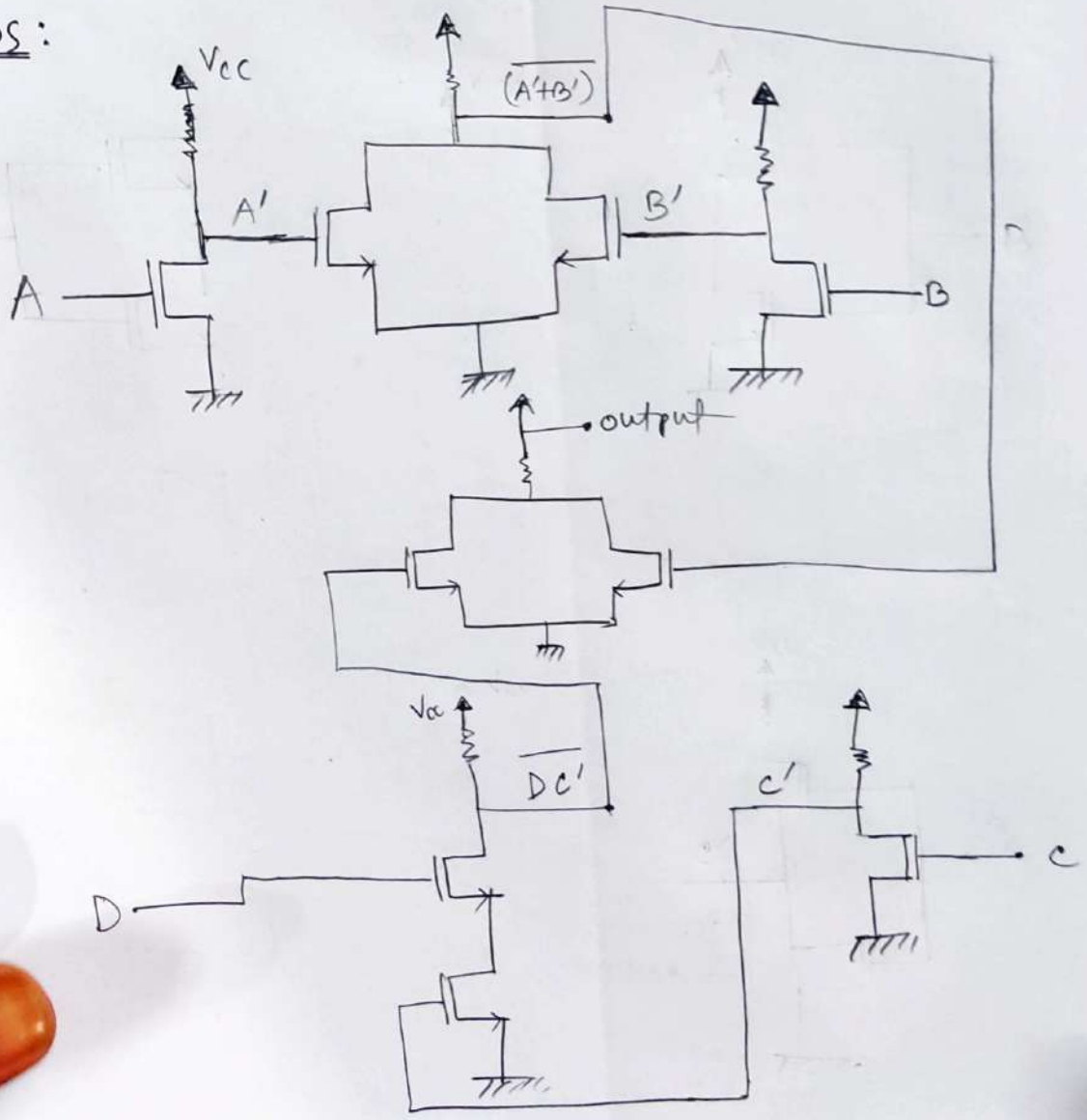
1. Design logic circuit using NMOS and DTL for the expression Y= (A'+B')DC'
2. Design logic circuit using CMOS for Y= (AB+C'D)'

① $Y = (A'+B')DC'$

$= \overline{(A'+B')DC'}$

$= \overline{(A'+B')} + \overline{DC'}$

NMOS:



AD BREAK

Download your cover page and exam routine, see Utsob's notes, and much more.

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22/9/2026

Mid এর আগে
Machine — DC Generator, Motor
electronics — Operational Amplifier

DC Generator

Faraday's laws (২টি)

Slide 2

Slide 4

$e \propto -\frac{d\phi}{dt}$ ← Farady's second law

$e = -N\frac{d\phi}{dt}$

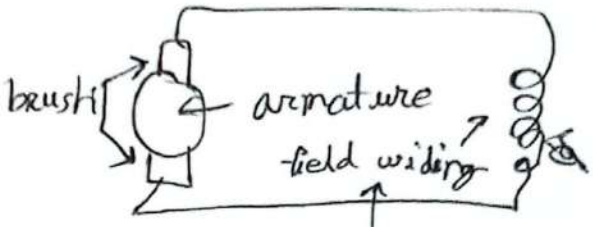
e = induced e.m.f
$\frac{d\phi}{dt}$ = rate of change of flux
N = number of turns of the coil

Lenz law

Slide 6

Generator AC generate করে
কোন generator directly DC generate করতে পারে না।
Split Ring — AC কে DC করতে ← commutator বলা হয়
Primary component
force
magnatic flux
current

ফ্যারাডের Law
লেঞ্জের Law



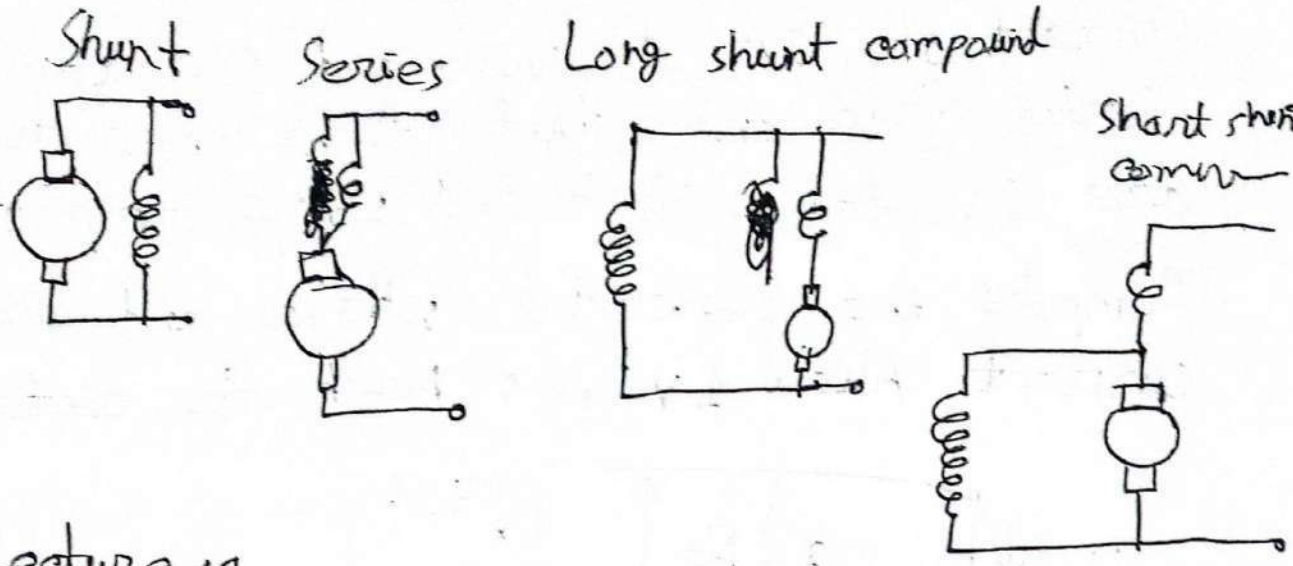
parallel এ shunt
Series এ series

Slide 11

Types of DC Generator

Circuit draw করা Mandatory

Assume 1V voltage drop per brush এবং 2টা brush
∴ Total 2V voltage drop



Lecture 14

Math 1

Example: A shunt generator delevers 450 A at 230 V and the resistance of the shunt field and armature are 50 Ω and 0.03 Ω respectively. Calculate the generated e.m.f.

Solution

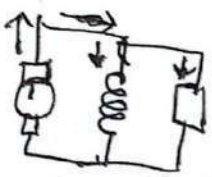
$$I_{sh} = \frac{V}{R} = \frac{230}{50} = 4.6A$$

$$I_a = I + I_{sh}$$
$$= 450 + 4.6 = 454.6A$$

Voltage drop $= I_a R_a = 454.6 \times 0.03 = 13.6V$

$$E_g = V_a + I_a R_a = 230 + 13.6 = 243.6V$$

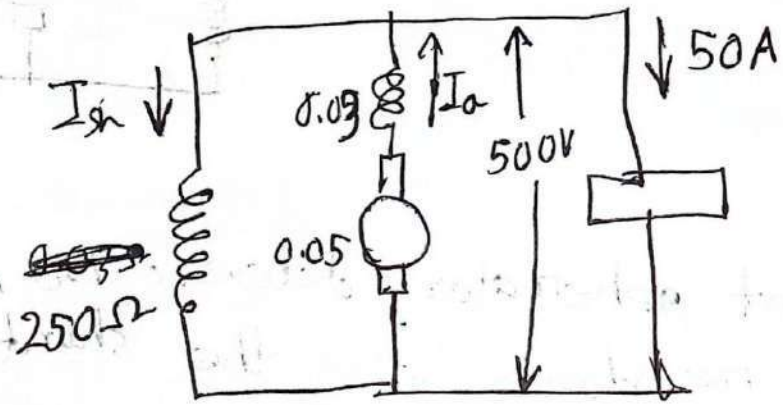
২টি slide এ আসা আছে



28/9/26

Lecture_DC-generator-1.pdf

[1] A long shunt compound generator delivers a load current of 50 A at 500 V and has armature, series field and shunt field resistances are 0.05 Ω, 0.03 Ω and 250 Ω respectively. Compute the generated voltage and armature current. Allow 1 V per brush for contact drop.



$$I_{sh} = \frac{500}{250} = 2A$$

$$I_a = I_{sh} + I = 2 + 50 = 52A$$

Voltage drop = $0.03 \times 52 + 0.05 \times 2$

$= 6.76 V = 1$

$E_g = 500 + 6.76 = 506.76 V$

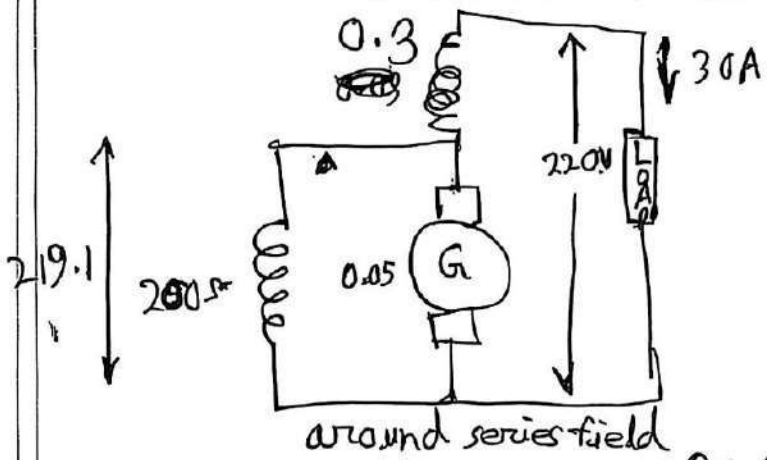
Voltage drop on series $= 52 \times 0.03 = 1.56 V$

armature drop $= 0.05 \times 52 = 2.6$

Drop at brushes $= 2 \times 1 = 2V$

$E_g = 500 + 2.6 + 1.56 + 2 = 506.16 V$

A short shunt compound generator delivers a load current of 30 A at 220 V and has armature, Series field and shunt field resistances are 0.05 Ω, 0.3 Ω and 200 Ω respecectively. Compute the generate voltage armature current. Allow 1 v per brush



• Voltage drop = 0.3 × 30 A = 9 V

∴ Voltage around armature = 220 + 9 = 229 V ~~219.1 V~~

$I_{sh} = \frac{229}{200} = 1.145$ A ~~1.0955~~

$I_a = I_{sh} + 30 = 1.145 + 30 = 31.145$ A ~~1.0955~~

$I_a R_a = 31.145 \times 0.05 = 1.56$ V

Brush drop = 2 × 1 = 2 V

DC Generator generated EMF equation

$$E_g = \frac{\phi Z N}{60} \times \frac{P}{A} \text{ volt}$$

Wave winding = Series
Lap winding = parallal

→ wave winding has only two parallal paths (Low current, High voltage)

Lap winding has multiple parallal paths (High current, low voltage).

ϕ = flux/pole in weber
Z = total no. of armature conductor
P = No of generator poles
A = No of parallal paths in armature
N = armature rotation in revolution per minute
E = EMF induced in any parallal path in armature

for wave winding $A = 2$
for lap winding $A = P$

next week Monday class test

A four pole generator having wave wound armature winding has 51 slots each slot containing 20 conductors. Calculate E_g, when driven at 1500 rpm assuming flux per pole is 7 mWb.

$$P = 4$$
$$A = 2$$
$$N = 1500$$
$$\phi = 7 \times 10^{-3}$$
$$Z = 51 \times 20 = 1020$$
$$E_g = \frac{7 \times 10^{-3} \times 1020 \times 1500}{60} \times \frac{4}{2}$$
$$= 357 \text{ V}$$

rps থেকে কিভাবে rpm বের করতে হয়

$$\text{rpm} = 60 \times \text{rps}$$

29/9/2026

An 8-Pole dc generator has 800 armature conductors and flux of 0.05 Wb/pole. Compute E_g if it is Lap connected and runs at 1200 rpm. What will be the speed at which it is to be drived to produce the same e.m.f if it is wave wound.

$P = 8$

$A = P = 8$

$N = 1200$

$\phi = 0.05 \times 8 = 0.4$

$Z = 800$ 0.05

$$E_g = \frac{0.4 \times 800 \times 1200}{60} \times \frac{8}{8} = 6400\ V$$

$$= 800\ V$$

if wave wound $A = 2$

$$800 = \frac{0.05 \times 800 \times N}{60} \times \frac{8}{2}$$

$$\Rightarrow N = 300\ \text{rpm}$$

1 Wb = 10^{-8} Maxwells (M_x)

Wb = $10^{-8} \times M_x$

An 6-pole dc generator has 800 armature conductors and flux of ~~0.05 Wb/pole~~ 27000 ~~Maxw~~ M_x/Pole ~~Compute E_g = 800 V if~~

Compute E_g if it is lap connected and runs at 70 r.p.s. What will be the speed at which it is to be drived to produce the same e.m.f if it is wave wound.

$P = 6$

$Z = 800$

$\phi = 27000 \times 10^{-8} = 2.7 \times 10^{-4}$ Wb

$A = P = 6$

$N = 70 \times 60 = 4200$ rpm

$$E_g = \frac{\phi Z N}{60} \times \frac{P}{A} = \frac{2.7 \times 10^{-4} \times 800 \times 4200}{60} \times \frac{6}{6}$$

$$= 15.12 \text{ V}$$

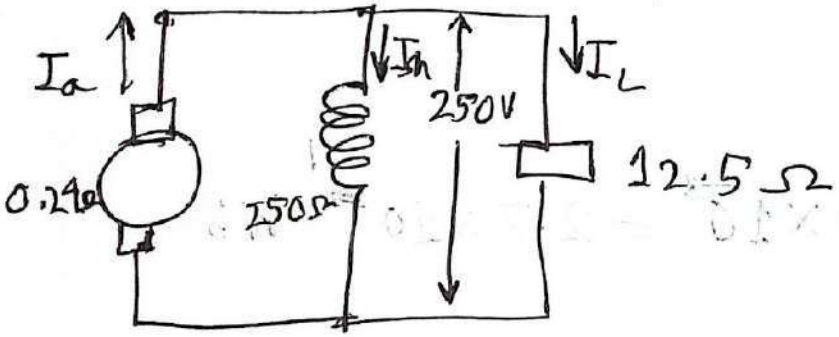
wave wond $A = 2$

$$15.12 = \frac{2.7 \times 10^{-4} \times 800 \times N}{60} \times \frac{6}{2}$$

$\Rightarrow N = 1400$ rpm

An 8 pole, dc Shunt gen with 778 wave Connected armature conductors and running at 500 r.p.m. Supplies a load of 12.5Ω resistance at terminal voltage of 250 v. The armature resistance is 0.24Ω and field resistance is 250Ω. Find armature current, induced e.m.f and flux per pole

↑ 9.83 mwb



$$I_{sh} = \frac{V}{R} = \frac{250}{250} = 1\ A$$

$$I_L = \frac{V}{R} = \frac{250}{12.5} = 20\ A$$

armature current $I_a = I_{sh} + I_L = 1 + 20 = 21\ A$

Voltage drop $= I_a R_a = 21 \times 0.24 = 5.04\ V$

$\therefore$ induced e.m.f $= 250 + 5.04$

$= 255.04\ V$

$E_g = 255.04$

$\phi = ?$

$P = 8$

$Z = 778$

$A = 2$

$N = 500$

$$255.04 = \frac{\phi' \times 778 \times 500}{60} \times \frac{8}{2}$$

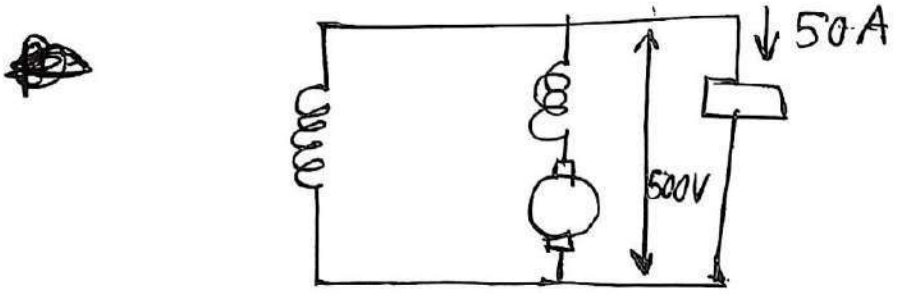
$$\Rightarrow \phi = 9.83 \times 10^{-3} \text{ Wb}$$

A Textbook of electrical technology } Machine related Topic
BL Theraja (Volume-II)

A 4 pole, long shunt, lap wound gen delivers load current, 50 A and voltage 500 V

The armature, series and shunt field resistances are 0.03 Ω, 0.04 Ω, 200 Ω. The brush drop is 1 V/brush. Determine induced voltage.

Calculate the no of conductors if the speed is 1200 rpm and flux per pole is 0.02 wb.



6/10/2026

DC Motor

Lecture 9_ DC Motor

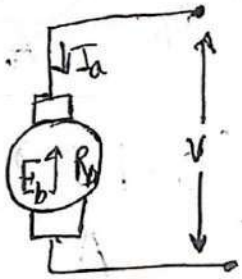
A DC motor is an electrical machine that converts electrical energy into mechanical energy

input – electrical
output - mechanical

Torque:

Torque is the measure of the force that can cause an object to ro_____

Flemings left hand rule (DC generator ✓ Right hand rule)



$V = E_b + I_a R_a$

E_b = back EMF

The direction of the induced emf is opposite to the applied voltage

$$E_b = \frac{NP\Phi Z}{60A}$$

E_b = the induced emf of motor
A = the number of parallel paths through the armature
P = The number of poles
N = The speed
Z = The total no of conductor

Φ = flux per pole

The back emf makes the DC motor, self regulating machine.

Torque and Speed Equation

Torque equation,

We know $V = E_b + I_a R_a$

$$V I_a = E_b I_a + I^2 R_a$$

Mechanical Power required to rotate the shaft $I\omega$; $\omega = \frac{2\pi N}{60}$ rps

So, $T\omega = E_b I_a$

Put the value of $E_b = \frac{N \phi Z P}{A 60}$

$$T = \left(\frac{0.159 P Z}{A}\right) \phi I_a$$

$\frac{1}{2\pi} = 0.159$

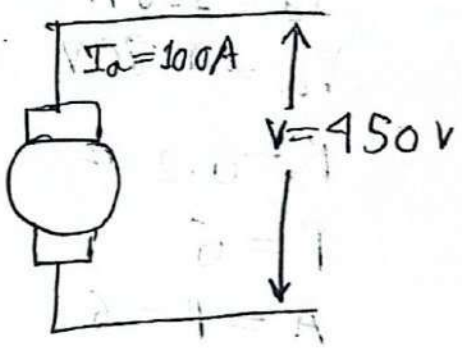
T = Torque (τ)

Slide 15/23

1. $I_a = 100A$

$V_a = 450V$

$R_a = 2.5\Omega$



$P = 6$

$A = 2$

$Z = 850$

$\phi = 0.05Wb$

$V = E_b + I_a R_a$

$450 = E_b + 100 \times 2.5$

$\Rightarrow E_b = 450 - 100 \times 2.5$

$= 200\ V$

$$E_b = \frac{N P \phi Z}{60 A}$$

$$200 = \frac{N \times 6 \times 0.05 \times 850}{60 \times 2}$$

$$\Rightarrow N = \frac{200 \times 60 \times 2}{6 \times 0.05 \times 850} = 94.12$$

$$T = \left(\frac{0.159 \times 6 \times 850}{2}\right) \times 0.05 \times 100$$

$$= 2027.25\ Nm$$

2. $I_a = 110\ A$

$V_a = 480\ V$

$R_a = 0.2\ \Omega$

$P = 6$

$A = P = 6$

$Z = 864$

$\phi = 0.05\ Wb$

$$V = E_b + I_a R_a$$

$$\Rightarrow 480 = E_b + 110 \times 0.2$$

$$\Rightarrow E_b = 480 - 110 \times 0.2 = 458\ V$$

$$E_b = \frac{NP\phi Z}{60A}$$

$$\Rightarrow 458 = \frac{N \times 6 \times 0.05 \times 864}{60 \times 6}$$

$$\Rightarrow N = \frac{458 \times 60 \times 6}{6 \times 0.05 \times 864} = 636.11$$

$$T = \left(\frac{0.159 PZ}{A}\right) \phi I_a$$

$$= \frac{0.159 \times}{}$$

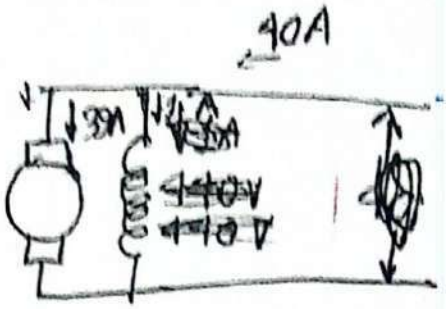
$P = 6$

$Z = 540$

$A = P = 6$

$\phi = 30\ mWb = 0.03\ Wb$

$R_a = 0.05\Omega$



$P_a = 8.595kW = 8595\ W$

$P_a = \frac{V_a^2}{R_a}$

$\Rightarrow V_a = \sqrt{P_a R_a} = \sqrt{8595 \times 0.05} = 20.73V$

~~$V_a = 20.73$~~
~~$R_a = 0.05$~~

$I_a = 39\ A$

$V = E_b + I_a R_a$

$20.73 = E_b + 39 \times 0.05$

$\Rightarrow E_b = 0.02667$

$E_b = \frac{NP\phi Z}{60A}$

$\Rightarrow 0.02667 = \frac{N \times 6 \times 0.03 \times 540}{60 \times 6}$

$\Rightarrow N = \frac{0.02667 \times 60 \times 6}{6 \times 0.03 \times 540} = 0.988$

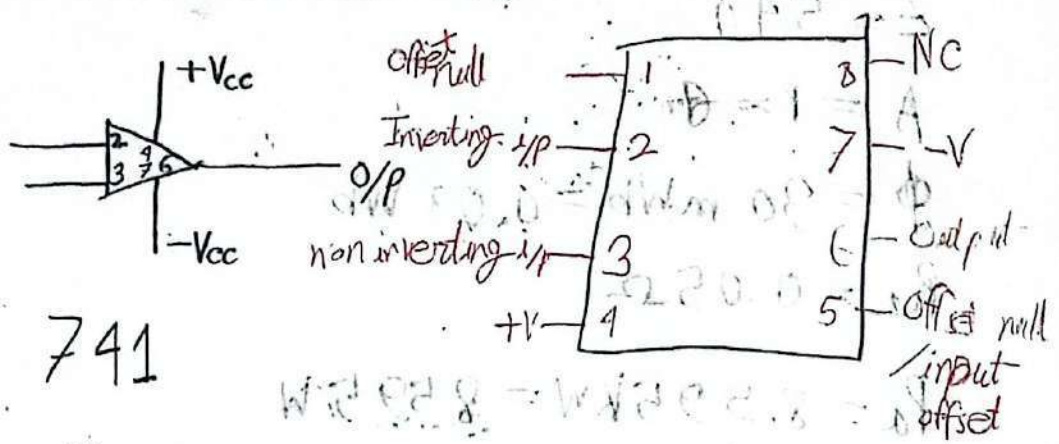
~~$\frac{NP\phi Z}{60A}$~~ $T = \left(\frac{0.159PZ}{A}\right)\phi I_a$

$= \frac{0.159 \times 6 \times 540}{6} \times 0.03 \times 39 = 37.276$

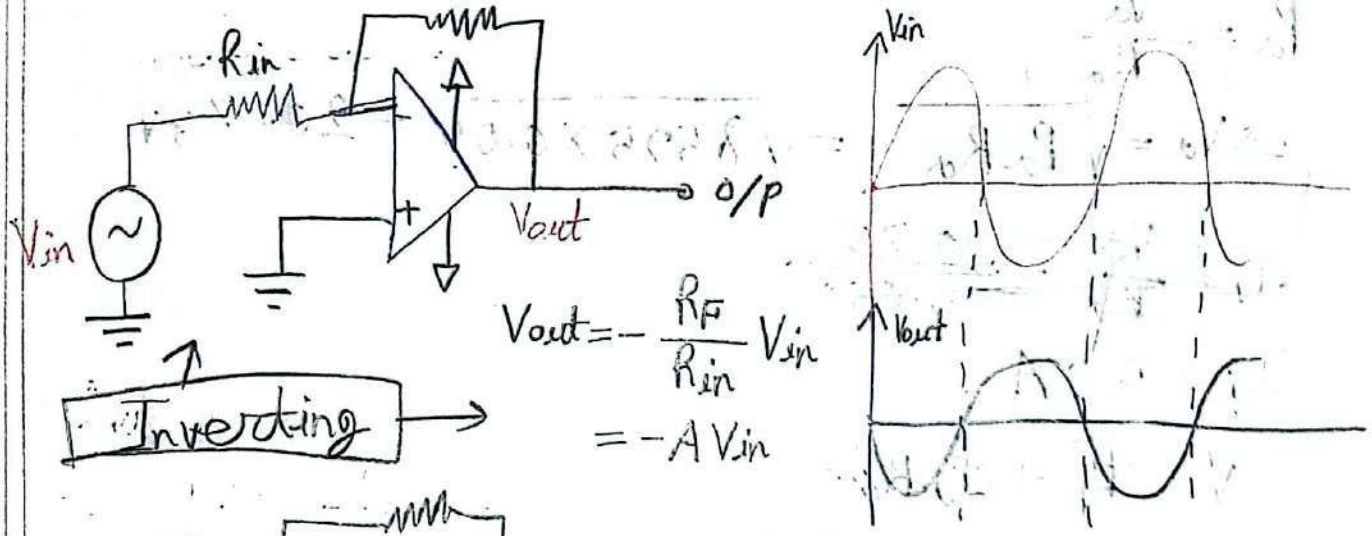
2/10/26

Mice sound ক Amplify করতে

Operational Amplifier



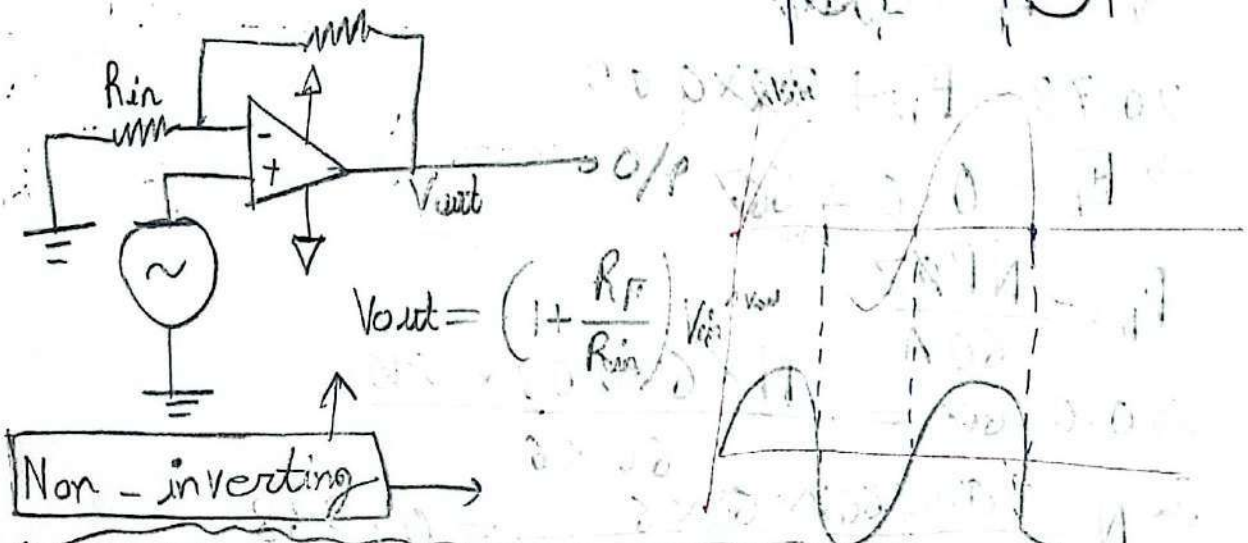
741



$$V_{out} = -\frac{R_F}{R_{in}} V_{in}$$

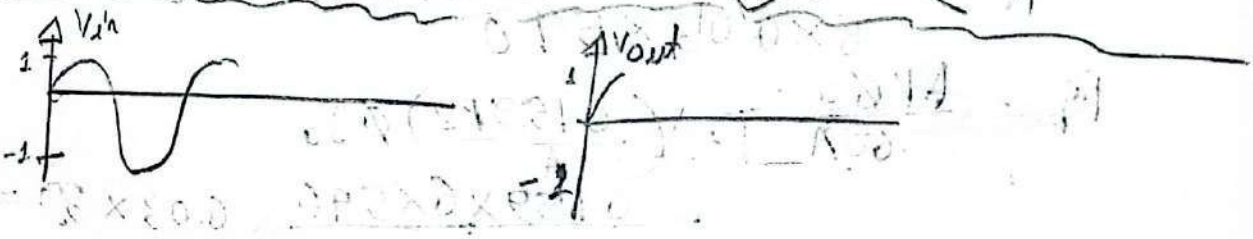
$$= -A V_{in}$$

Inverting →

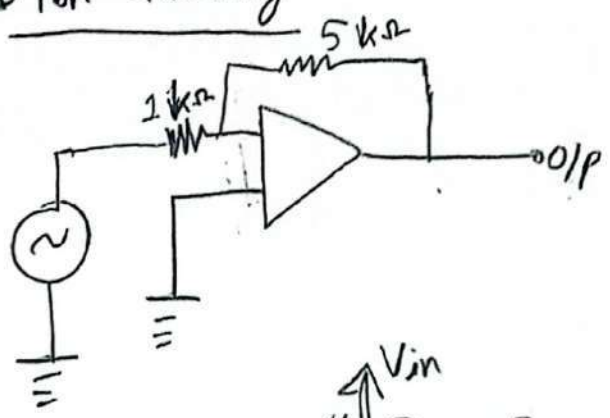


$$V_{out} = \left(1 + \frac{R_F}{R_{in}}\right) V_{in}$$

Non - inverting →



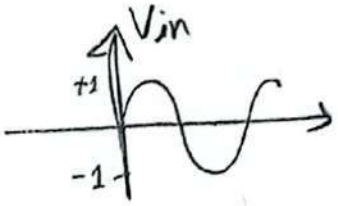
Non-inverting

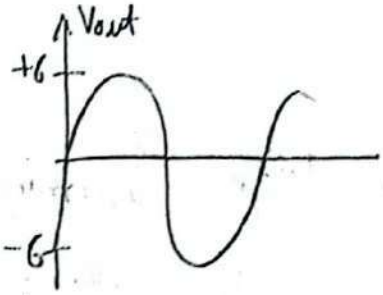


$$V_{out} = \left(1 + \frac{R_F}{R_{in}}\right) V_{in}$$

$$= \left(1 + \frac{5}{1}\right) \times 1$$

$$= 6\ V$$



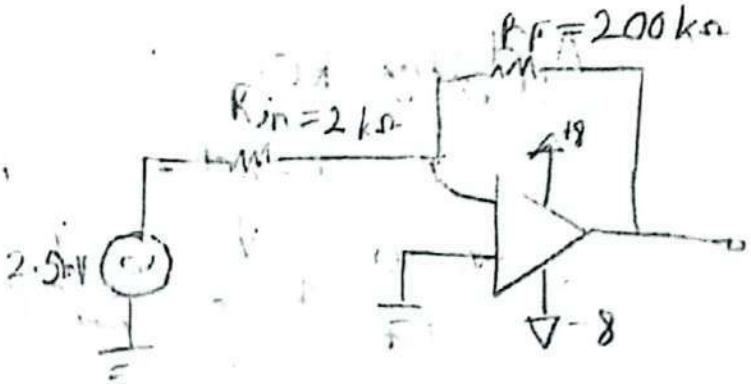


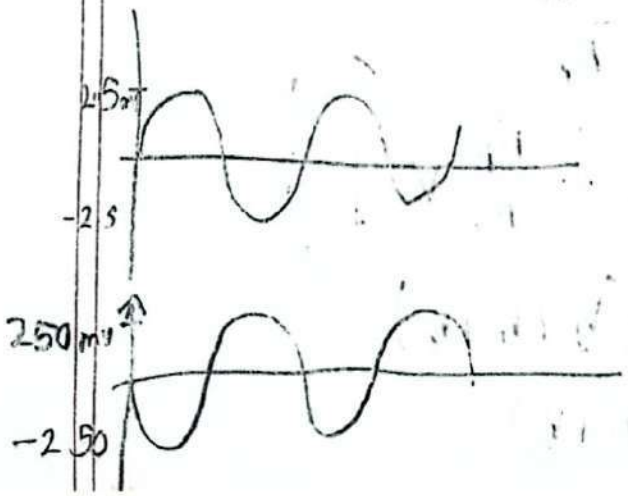
Inverting

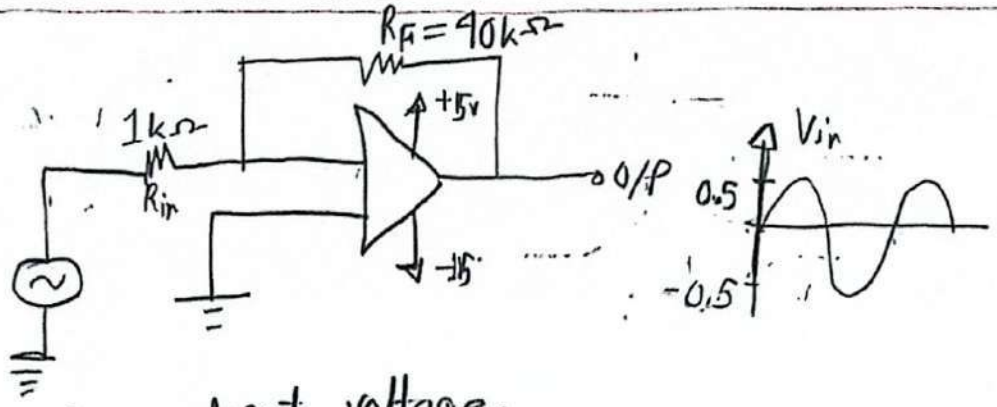
$$V_{out} = -\frac{R_F}{R_{in}} \times V_{in}$$

$$= -\frac{200}{2} \times 2.5$$

$$= -250\ mV$$



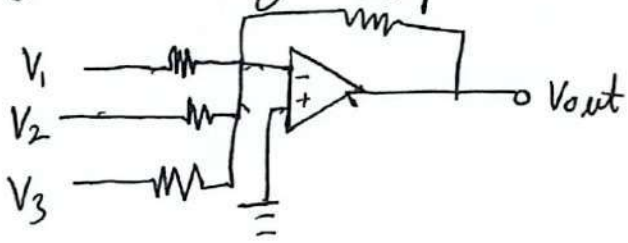




Determin output voltage.

$$V_{out} = -\frac{R_F}{R_{in}} \times V_{in} = -\frac{40}{1} \times 0.5 = -20\,V$$

Inverting Summing amplifier



Applying KCL,

$$I_1 + I_2 + I_3 = I_F$$

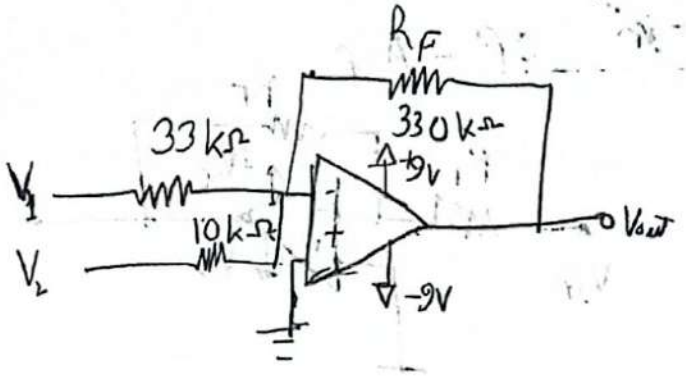
$$\Rightarrow \frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} = \frac{0 - V_{out}}{R_F}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_{out}}{R_F}$$

$$\Rightarrow V_{out} = -\left[\frac{R_F}{R_1} V_1 + \frac{R_F}{R_2} V_2 + \frac{R_F}{R_3} V_3\right]$$

If, $R_1 = R_2 = R_3$, $V_{out} = -\frac{R_F}{R}(V_1 + V_2 + V_3)$

If $R_F = R$, $V_{out} = -(V_1 + V_2 + V_3)$



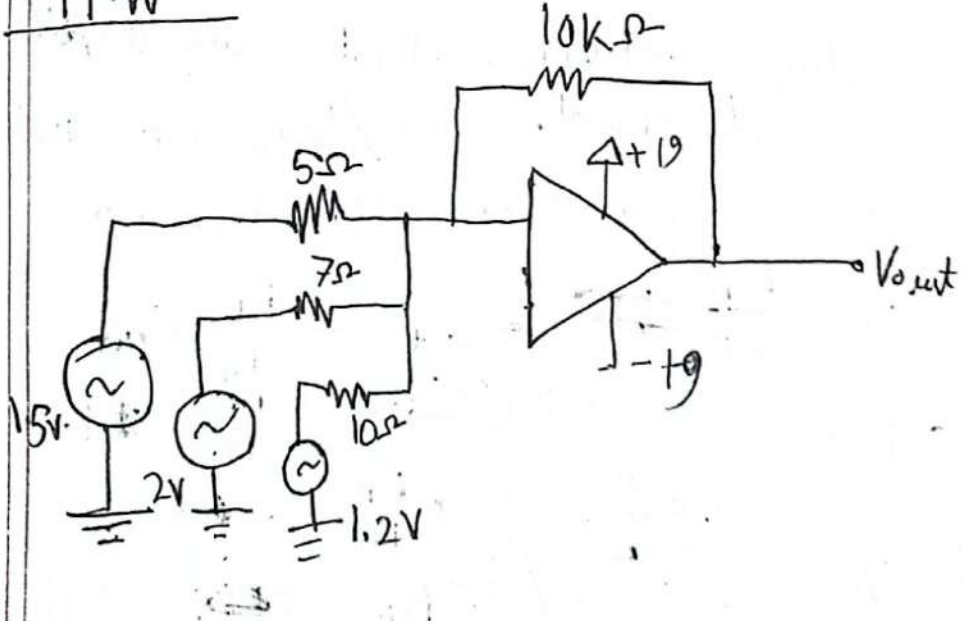
$v_1 = 50\sin(100t)$ mV

$v_2 = 10\sin(300t)$ mV

$$V_{out} = -\left[\frac{330}{33} \times 50\sin(100t) + \frac{330}{10} \times 10\sin(300t)\right]$$

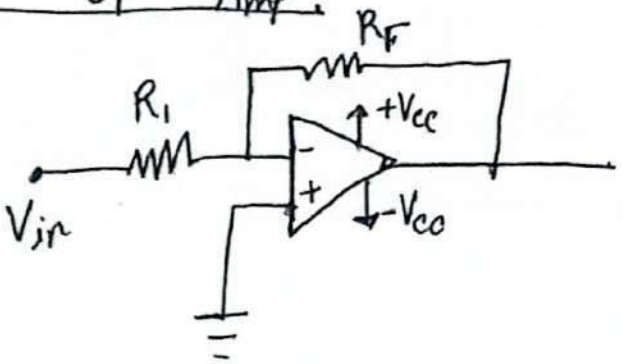
$$= -1500\sin(100t) - 330\sin(300t) \text{ mV}$$

H.W



Virtual ground in op-Amp.

$$A_V = \frac{V_{out}}{V_{in}}$$



$$V_{out} = A_V \times V_{in}$$

$$= A_V \times V_d$$

$$5 = 10^5 \times V_d$$

$$V_d = 0.00005$$

$$V_d \approx 0$$

$$V^+ - V^- = 0; \quad V^+ = V^-$$

i/p = input
o/p = output

Develop an inverting amplifier that can amplify 5 times of input signal. Show/draw i/p and o/p waveforms. Consider a 3V(P-P) as i/p.

Ans:

$$V_{out} = -\frac{R_F}{R_1} V_{in}$$

$$= -\frac{10\,k\Omega}{2\,k\Omega} \times 1.5$$

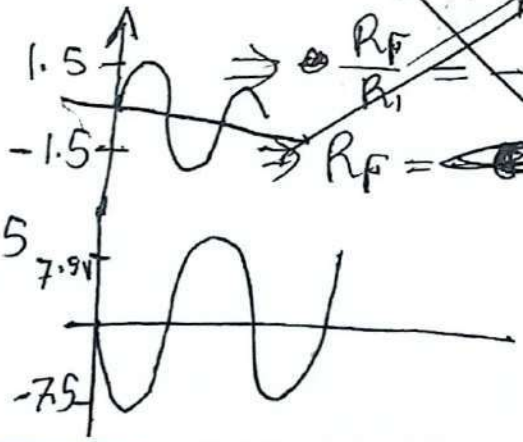
$$= -7.5\,V$$

$$V_{out} = -\frac{R_F}{R_1} \times V_{in}$$

$$1.5 \times 5 = -\frac{R_F}{R_1} \times 1.5$$

$$\frac{R_F}{R_1} = -5$$

$$R_F = -5R_1$$

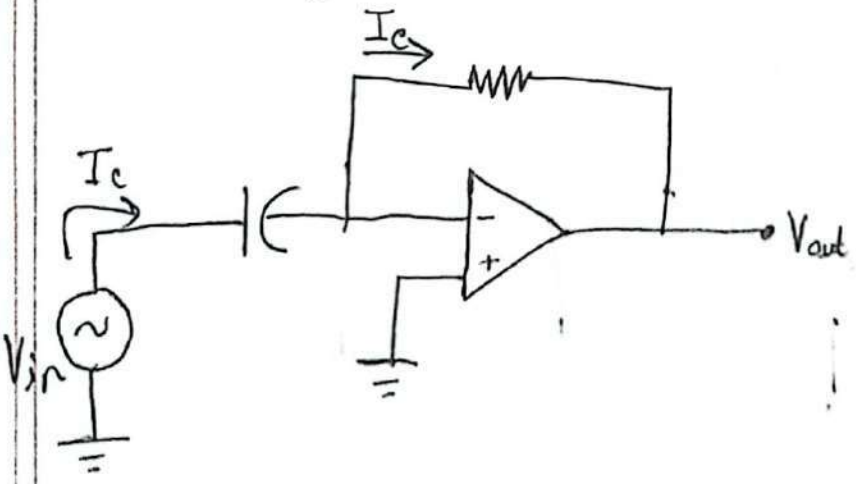


Slew rate

Slew rate is defined as the maximum rate of change of o/p voltage per unit time (V/s). How fast the op-amp is able to reach its final value / How fast the o/p voltage can change

OP-amp differentiator

$V_{in}(t) \rightarrow$ [Differentiator] $\rightarrow V_{out} = \frac{dV_{in}}{dt}$

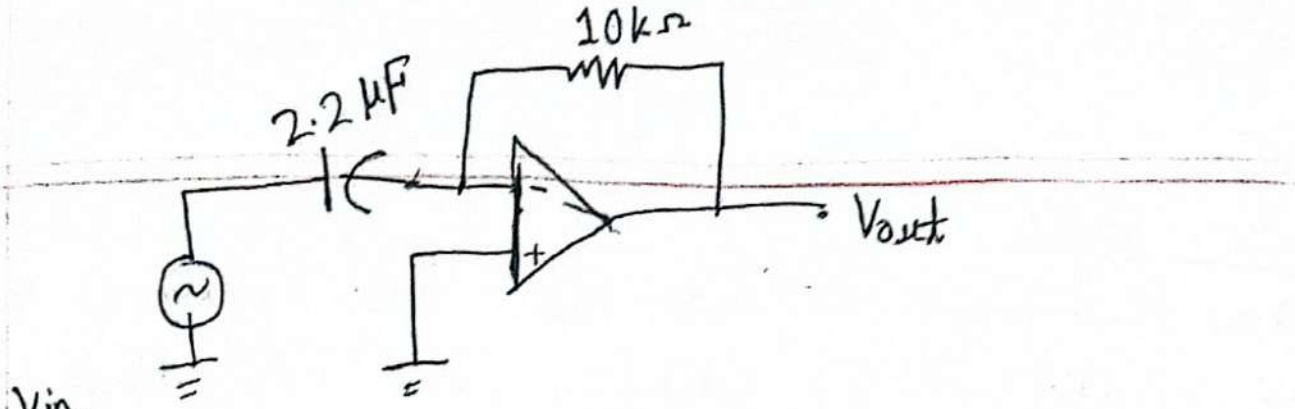


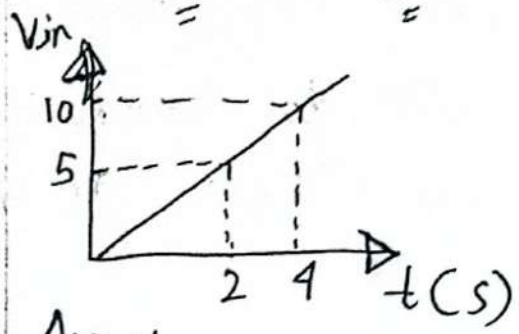
$I_c = I_f$

$\Rightarrow C\frac{dV_c}{dt} = \frac{0 - V_{out}}{R_F}$

$\Rightarrow C\frac{dV_{in}}{dt} = -\frac{V_{out}}{R_F}$

$\Rightarrow V_{out} = -R_F C\frac{dV_{in}}{dt}$





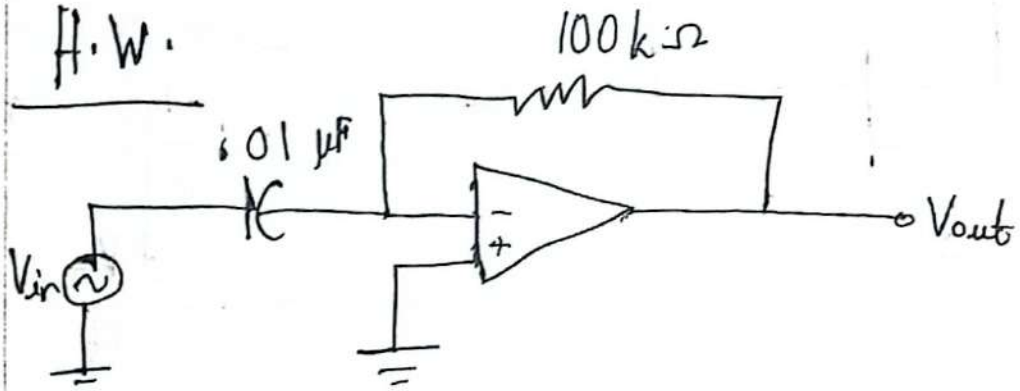
Ans:

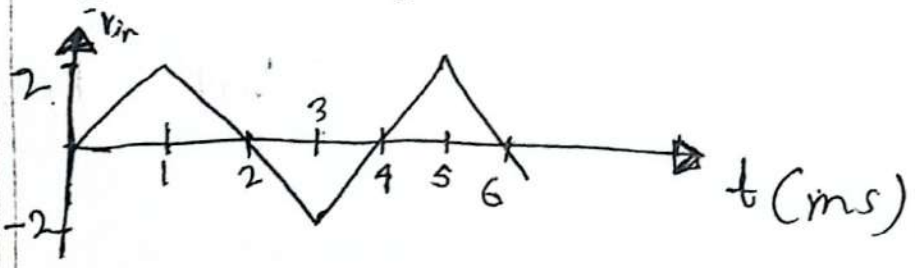
$$V_{out} = -R_F C \frac{dV_{in}}{dt}$$

$$\Rightarrow V_{out} = -10000 \times 2.2 \times 10 \times 10^{-6} \times \frac{5-0}{2-0}$$

$$= -0.55\ V$$

H.W.





Plot the o/p waveform of the op-amp differentiator given ~~below~~ up.

Mid + Ct (1+2+3) = 17 + (9+9+9)

next class op-amp o integrator

1. A DC Shunt Motor is connected to a 240V Supply. It is running at a speed of 1500 r.p.m. having 0.68Ω armature resistance and 40A armature current. Friction an windage loss in ~~600W~~ 600W. Find the ~~shaft~~ shaft torque.

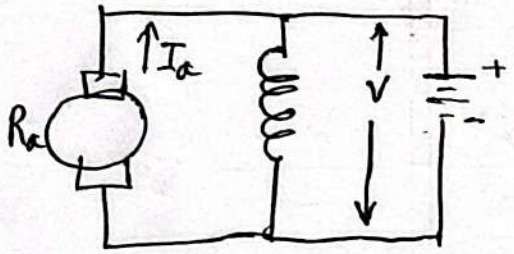
2. A dc shunt motor produce 250V back emf at 1500 r.p.m. If 50A current flows through the machine compute the electrical power converted to mechanical power and torque?

(1)

$V = 240V$, $N = 1500 rpm$, $R_a = 0.68\Omega$, $I_a = 40A$

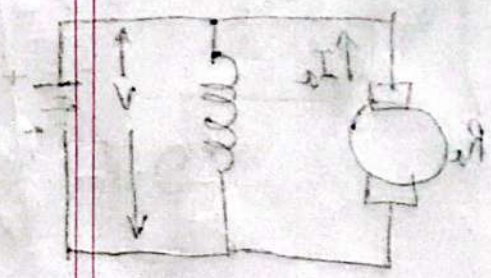
Friction and Windage loss = 600 W

$V = I_a R_a +$

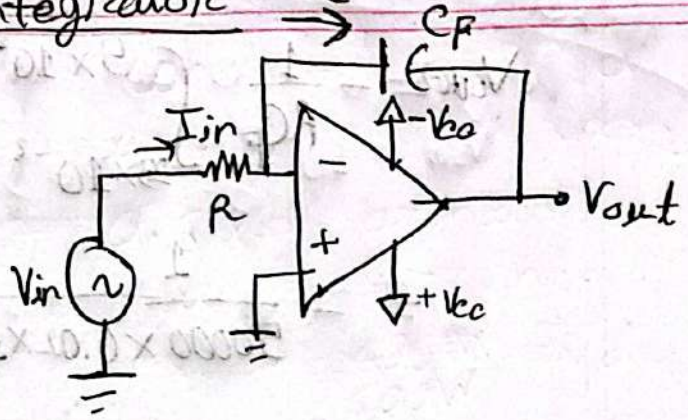


এটা একটা সাদা পেইজ

(1)



OP Amp Integrator



$$I_{in} = I_c$$

$$\frac{V_{in} = V_x}{R} = C_F \frac{\partial V_c}{\partial t}$$

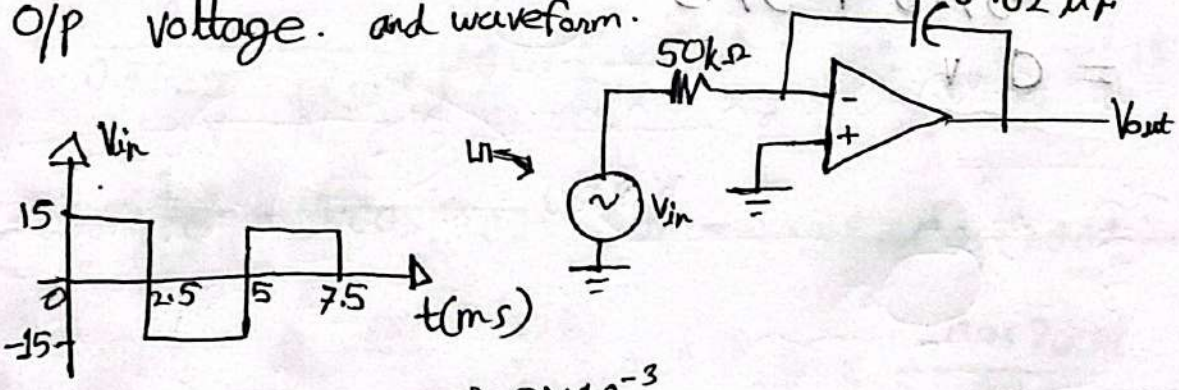
$$\Rightarrow \frac{V_{in} - 0}{R} = C_F \frac{\partial}{\partial t}(V_x - V_{out})$$

$$\Rightarrow \frac{V_{in}}{R} = C_F \frac{\partial}{\partial t}(0 - V_{out})$$

$$\Rightarrow V_{out} = -\frac{1}{RC_F} V_{in}$$

After integration, $\boxed{V_{out}(t) = -\frac{1}{RC_F}\int V_{in}(t)}$

For 200 Hz rectangular pulse, determine the O/P voltage. and waveform.



$$V_{out} = \frac{-1}{RC_F}\int_0^{2.5\times10^{-3}} 15\,dt$$

$$= \frac{-1}{50000 \times 0.02\times10^{-6}} \times [15t]_0^{2.5\times10^{-3}}$$

$$= \frac{-1}{50000\times0.02\times10^{-6}} \times 15\times2.5\times10^{-3} = -37.5\ V$$

$$V_{out}(t) = -\frac{1}{RC_F}\int_{25\times10^{-3}}^{5\times10^{-3}} -15\,dt$$

$$\Rightarrow -\frac{1}{50000\times0.02\times10^{-6}}\left[-15t\right]_{25\times10^{-3}}^{5\times10^{-3}}$$

~~37.5~~

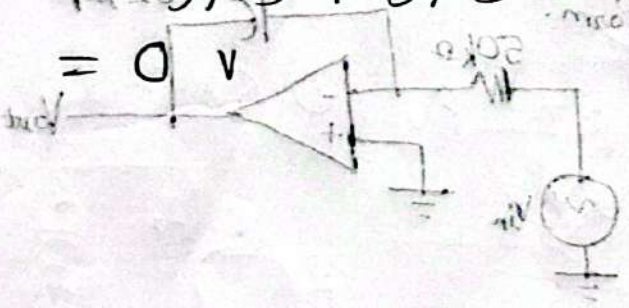
$$= -\frac{1}{50000\times0.02\times10^{-6}}\times(-15\times5\times10^{-3}+15\times2.5\times10^{-3})$$

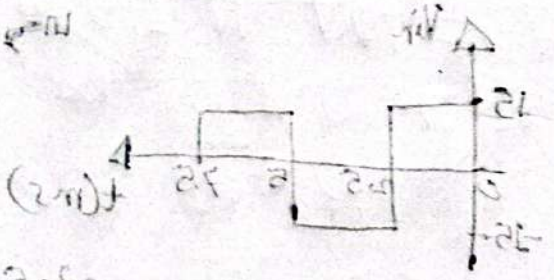
$$= 37.5$$

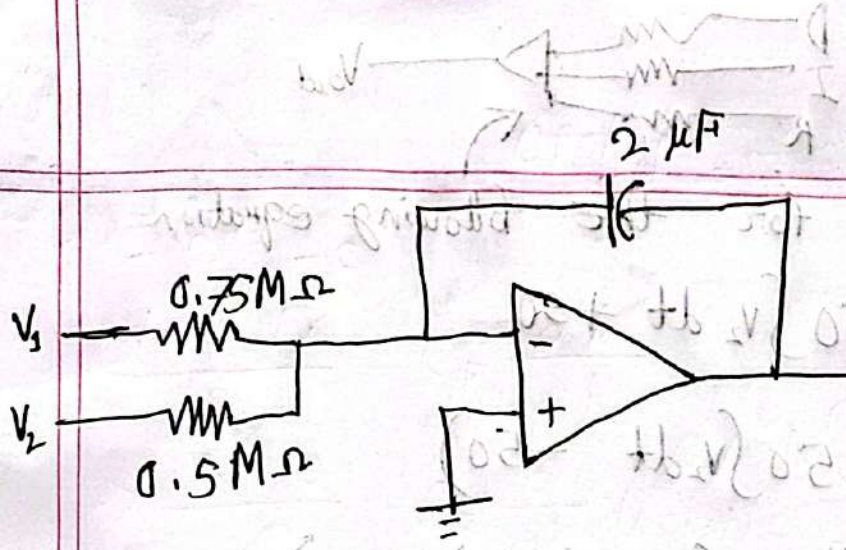
$$\therefore V_o = V_{out}(+) + V_{out}(-)$$

$$= -37.5 + 37.5$$

$$= 0\ V$$







$V_1 = 10 \sin(200t)$

$V_2 = 15 \sin(200t)$

Compute $V_{out} = ?$

~~$V_o = -\frac{1}{0.75 \times 10^6 \times 2 \times 10^{-6}} \int 10 \sin(200t)\, dt$~~

$$V_o = -\frac{1}{0.75 \times 10^6 \times 2 \times 10^{-6}} \int 10 \sin(200t)\, dt - \frac{1}{0.5 \times 10^6 \times 2 \times 10^{-6}} \int 15 \sin(200t)\, dt$$

~~$= -\frac{1}{1.5} \times 10 \cos(200t) \times 200 - 1 \times 15 \times \cos(200t)$~~

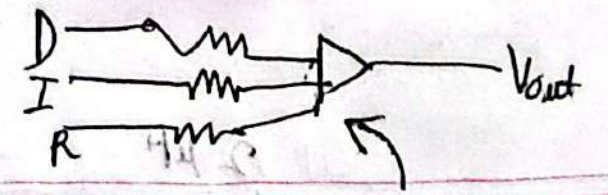
$$= -\frac{1}{1.5} \times 10 \frac{-\cos(200t)}{200} - 1 \times 15 \times \frac{-\cos 200t}{200}$$

$$= 0.033 \cos(200t) + 0.075 \cos(200t)$$

$$= 0.1083 \cos(200t) \text{ volt}$$

Scanned with CamScanner

Practice Problem



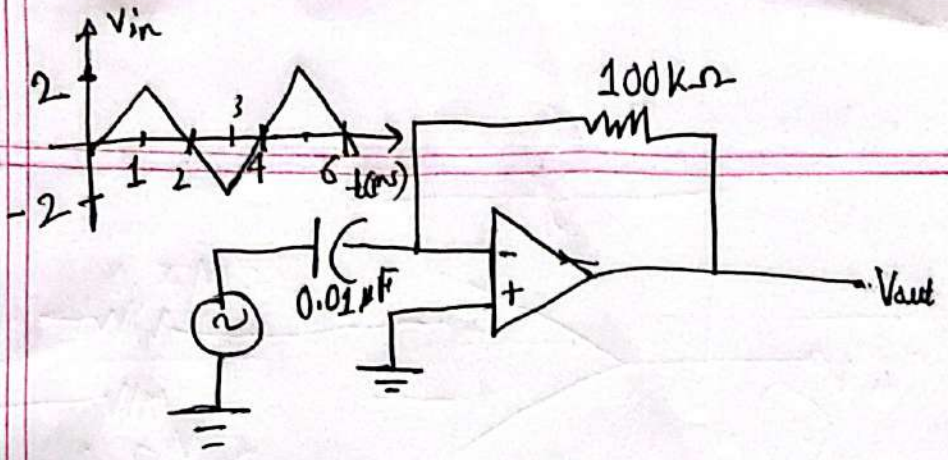
Design a circuit for the following equation

$$V_{out} = 10\frac{\partial V_1}{\partial t} - 50\int V_2\,dt + 50$$

$$= -\left(-10\frac{\partial V_1}{\partial t} + 50\int V_2\,dt - 50\right)$$

$$= -\left(-10\frac{\partial V_1}{\partial t} - \left(-50\int V_2\,dt\right) - 50\right)$$

D I R



$T = 4$

$y = mx + c$

$m = \frac{y}{x} = \frac{y_1 - y_2}{x_1 x - x_2} = \frac{0-2}{0-1} = 2$

$0 < t < 1 \longrightarrow y = mx$ (মূলবিন্দুগামী)

$= 2 \times t = 2t$

$1 < t < 0$

$3 < t < 4$

$y = mx + c$

$= -2t + 4$

$m = \frac{y_1 - y_2}{x_1 - x_2} = \frac{2-(-2)}{1-3} = -2$

(1,2) বিন্দুগামী

$2 = -2 \times 1 + c$

$\Rightarrow c = 4$

$y = mx + c$

$\therefore y = 2t - 8$

$m = \frac{y_1 - y_2}{x_1 x_2} = \frac{-2-0}{3-4} = 2$

$0 = 2 \times 4 + c$

$\Rightarrow c = -8$

$Vin = 2t + (-2t + 4) + (2t - 8)$

$= 2t - 4$

$\therefore Vout = -R_F C \frac{dVin}{dt} = -100000 \times 0.01 \times 10^{-6} \times \frac{d}{dt}(2t-4)^2$

$= -2 \times 10^{-3}$ V